

JESS Thermodynamic Database v8.9

Titanium

24-Jan-23

Reaction No. 2692							
F-1 + TiO+2 = TiO+2_F-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.2	HCl	lgK:6.65 (3SF)	3	MCE	931
2	25	3	NaClO4	lgK:5.38 (3SF)	3	MQH	437

Reaction No. 2693							
TiO+2_F-1 + F-1 = TiO+2_F-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.2	HCl	lgK:5.09 (3SF)	3	MCE	931
2	25	3	NaClO4	lgK:4.32 (3SF)	3	MQH	437

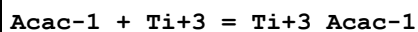
Reaction No. 2694							
TiO+2_F-1(2) + F-1 = TiO+2_F-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.2	HCl	lgK:4.58 (3SF)	3	MCE	931
2	25	3	NaClO4	lgK:3.96 (3SF)	3	MQH	437

Reaction No. 2695							
TiO+2_F-1(3) + F-1 = TiO+2_F-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.2	HCl	lgK:4.06 (3SF)	3	MCE	931
2	25	3	NaClO4	lgK:3.72 (3SF)	3	MQH	437

Reaction No. 4192							
EDTA-4 + TiO+2 = TiO+2_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:22	1	NaClO4	lgK:18.15 (0.02SD)	5	MSP	906
2	20	0.1	KClO4	lgK:17.5 (0.1SD)	7	EIU	903
3	20	0.1	KClO4	lgK:17.5 (0.1SD)	5	MDS	2373
4	23	0.5	H2SO4	lgK:17.3 (3SF)	3	MPL	903
5	25	0	Inf. Dilution	lgK:18.99 (4SF)	1	EOR	

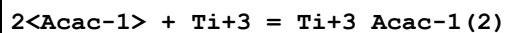
6	25	0	Inf. Dilution	lgK:19.0 (3SF)	2	EAE	
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Reaction No. 4536



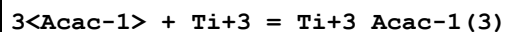
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:10.43 (4SF)	6	MSP	999

Reaction No. 4537



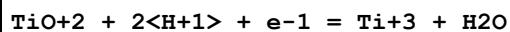
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:18.82 (4SF)	6	MSP	999

Reaction No. 4538



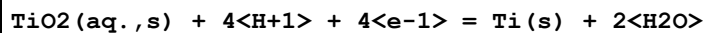
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.9 (3SF)	5	MGL	2360
2	25	1	Unknown	lgK:24.9 (3SF)	4	MSP	999

Reaction No. 4882



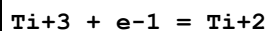
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.10 (2SF)	2	CRV	18118
2	25	0	Inf. Dilution	E:0.1 (1SF)	3	CRV	22551
3	25	0.1:3	HCl	E:0.03 (1SF)	0	MPT	20357
4	25	3	(H,Na)Cl	E:0.009 (0.002SD)	7	MEF	23226

Reaction No. 4884



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-60.8 (3SF)	1	ELC	

Reaction No. 4885



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-6.2 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-15.2 (0.05SD)	0	ENS	802
3	25	0	Inf. Dilution	E:-0.9 (1SF)	0	CRV	6330

4	25	0	Inf. Dilution	E:-0.37(0.01SD)	0	CRV	18118
5	25	0	Inf. Dilution	E:-1.63(3SF)	0	EFE	20358
Note (5): An 'estimate'; authors note as "about 0.4V more neg" than traditional							
6	25	0	Inf. Dilution	E:-0.9(1SF)	0	CRV	22519
7	25	0	Inf. Dilution	E:-2.3(2SF)	0	CRV	22551
Note (7): Values given are -2.3V or -0.37V! said to be due to irreversibility							

Reaction No. 4886							
Ti+2 + 2<e-1> = Ti(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-55.1(0.05SD)	3	ENS	778
2	25	0	Inf. Dilution	dG:314.(3SF)kJ	3	EFE	19339
3	25	0	Inf. Dilution	E:-1.630(4SF)	3	CRV	6330
4	25	0	Inf. Dilution	E:-1.25(3SF)	0	EOD	20358
Note (4): Low wgt for internal consistency							
5	25	0	Inf. Dilution	E:-1.630(4SF)	4	CRV	22519
6	25	0	Inf. Dilution	E:-1.628(4SF)	3	CRV	22551

Reaction No. 5746							
Ser-1 + Ti+3 = Ti+3_Ser-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:7.60(3SF)w	3	MPH	906
2	25	0.02	Unknown	lgK:7.54(3SF)	5	MGL	905

Reaction No. 8241							
Gly-1 + Ti+3 + H+1 = Ti+3_H+1_Gly-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.56(4SF)	1	EOR	
2	25	1	KCl	lgK:11.4(0.04SD)	4	MGL	1021

Reaction No. 8242							
Gly-1 + Ti+3 = Ti+3_Gly-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:8.50(3SF)w	3	MPH	906
2	25	0	Inf. Dilution	lgK:10.2(4SF)	1	EOR	
3	25	1	KCl	lgK:9.30(0.01SD)	6	MGL	1021

Reaction No. 8243							
Malonic-2 + Ti+3 = Ti+3_Malonic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	0.5	LiCl	lgK:6.53 (3SF) w	2	MPH	1702
2	15	0.5	LiCl	lgK:5.46 (3SF)	2	MSP	1702
3	25	1	KCl	lgK:6.8 (0.04SD)	3	MGL	1021

Reaction No. 8244							
Ti+3_Malonic-2 + Malonic-2 = Ti+3_Malonic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KCl	lgK:4.99 (0.02SD)	4	MGL	1021

Reaction No. 8245							
Ti+3_Malonic-2(2) + Malonic-2 = Ti+3_Malonic-2(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KCl	lgK:2.8 (0.03SD)	4	MGL	1021

Reaction No. 9196							
Ti+3 + H+1_NTA-3 = Ti+3_H+1_NTA-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1.2	KCl	lgK:18.7 (0.1SD)	3	MSP	1118
2	25	0	Inf. Dilution	lgK:18.9 (3SF)	1	ESC	

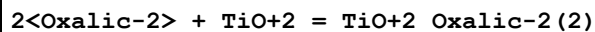
Reaction No. 12830							
EDDA-2 + Ti+3 = Ti+3_EDDA-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:8.75 (3SF)	6	MGL	2028

Reaction No. 12831							
Ti+3_EDDA-2 + EDDA-2 = Ti+3_EDDA-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:7.64 (3SF)	6	MGL	2028

Reaction No. 18346							
Oxalic-2 + TiO+2 = TiO+2_Oxalic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:6.60 (3SF)	0	ENS	1539

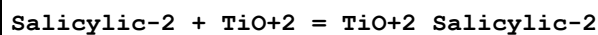
2	23	0.02	HClO ₄	lgK:6.6(0.1SD)	0	MSP	931
3	25	0	Inf. Dilution	lgK:10.96(4SF)	1	ELC	
4	25	0.03	Unknown	lgK:9.7(2SF)	0	MGL	931
5	25	0.5	NaClO ₄	lgK:7.90(0.02MD)	0	MGL	2638

Reaction No. 18347



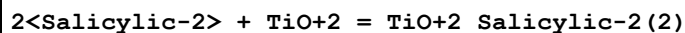
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:10.70(4SF)	0	ENS	1539
2	25	0	Inf. Dilution	lgK:14.42(4SF)	1	EOR	
3	25	0.5	NaClO ₄	lgK:13.24(0.07MD)	4	MGL	2638

Reaction No. 18348



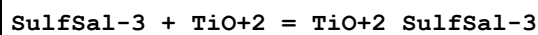
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:15.66(4SF)	6	ENS	1539
2	25	0	Inf. Dilution	lgK:16.5(3SF)	2	EAE	

Reaction No. 18349



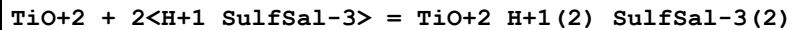
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:24.40(4SF)	6	ENS	1539
2	25	0	Inf. Dilution	lgK:25.2(3SF)	2	EAE	

Reaction No. 18350



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:3.10(3SF)	6	ENS	1539
2	25	0	Inf. Dilution	lgK:3.1(2SF)	1	ESC	

Reaction No. 18351



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:5.40(3SF)	6	ENS	1539
2	25	0	Inf. Dilution	lgK:5.5(2SF)	1	ESC	

Reaction No. 18352

TiO+2 + SulfSal-3 + 2<H+1_SulfSal-3> = TiO+2_H+1(2)_SulfSal-3(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:18.46(4SF)	6	ENS	1539
2	25	0	Inf. Dilution	lgK:18.8(3SF)	1	ESC	

Reaction No. 18353							
2<Tartaric-2> + TiO+2 = TiO+2_Tartaric-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:9.70(3SF)	6	ENS	1539

Reaction No. 18354							
TiO+2 + 2<H+1_Tiron-4> = TiO+2_H+1(2)_Tiron-4(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:15.00(4SF)	6	ENS	1539

Reaction No. 18355							
TiO+2 + Tiron-4 + 2<H+1_Tiron-4> = TiO+2_H+1(2)_Tiron-4(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:32.36(4SF)	6	ENS	1539

Reaction No. 23002							
Picolinic-1 + Ti+3 = Ti+3_Picolinic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	Br-1 salt	lgK:5.62(3SF)	6	MGL	999

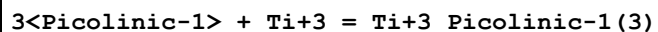
Reaction No. 23003							
Ti+3_Picolinic-1 + Picolinic-1 = Ti+3_Picolinic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	Br-1 salt	lgK:5.48(3SF)	6	MGL	999

Reaction No. 23004							
Ti+3_Picolinic-1(2) + Picolinic-1 = Ti+3_Picolinic-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	Br-1 salt	lgK:5.48(3SF)	6	MGL	999

Reaction No. 23005							
2<Picolinic-1> + Ti+3 = Ti+3_Picolinic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

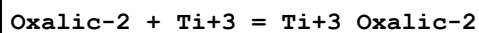
1	25	3	Br-1 salt	lgK:11.10 (4SF)	6	MSC	999
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Reaction No. 23006



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	Br-1 salt	lgK:16.58 (4SF)	6	MSC	999

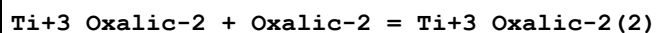
Reaction No. 26831



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	1	NaCl	lgK:6.45 (3SF)	2	MGL	1634
2	17	1.08	NaClO4	lgK:7.15 (3SF)	3	MIX	20376
3	25	0	Inf. Dilution	lgK:3.394 (4SF)	1	EOR	
4	25	1	Unknown	lgK:5.93 (3SF)	3	MSP	11516

Note (4): Lu K et al., Inorg. Chem., 1988, 27, 2325

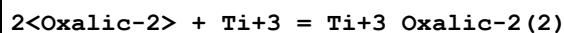
Reaction No. 26832



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	1	NaCl	lgK:6.08 (3SF)	2	MGL	1634
2	17	1.08	NaClO4	lgK:5.79 (3SF)	3	MIX	20376
3	25	1	Unknown	lgK:5.4 (2SF)	3	MSP	11516

Note (3): Lu K et al., Inorg. Chem., 1988, 27, 2325

Reaction No. 26833



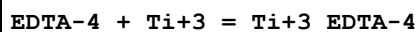
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	1	NaCl	lgK:12.53 (4SF)	5	MSP	1634
2	17	1.08	NaClO4	lgK:12.94 (4SF)	5	MIX	20376
3	25	0	Inf. Dilution	lgK:9.428 (4SF)	1	EOR	
4	25	0.1	KCl	lgK:8.58 (3SF)	3	MGL	906
5	25	0.2	KCl	lgK:8.47 (3SF)	3	MGL	906
6	25	0.2	NaNO3	lgK:12.11 (4SF)	0	MVM	11516

Note (6): Maluka L et al., Zhur. Neorg. Khim., 1980, 25, 1809

7	25	0.3	KCl	lgK:8.40 (3SF)	3	MGL	906
8	25	0.4	KCl	lgK:8.33 (3SF)	0	MGL	906
9	25	1	Unknown	lgK:11.33 (4SF)	3	MSP	11516

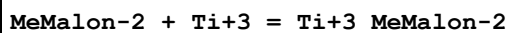
Note (9): Lu K et al., Inorg. Chem., 1988, 27, 2325

Reaction No. 29105



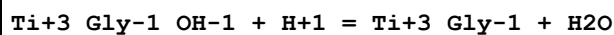
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	ClO4-1 salt	lgK:21.5 (3SF)	5	ENS	2373
2	22	0.1	Unknown	lgK:18.7 (3SF)	0	MPL	903
3	25	0	Inf. Dilution	lgK:23.82 (4SF)	1	EOR	
4	25	0.1	NaNO3	lgK:21.3 (3SF)	4	MPL	140
5	25			lgK:21.5 (3SF)	6	EIU	903

Reaction No. 30797



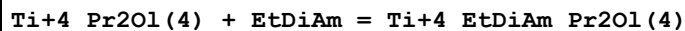
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	0.5	LiCl	lgK:6.60 (3SF)w	4	MPH	1702

Reaction No. 31071



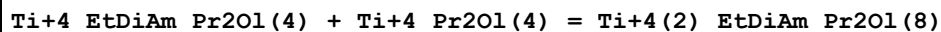
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:4.82 (3SF)	5	CRV	2556

Reaction No. 34364



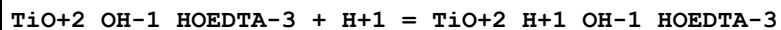
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	5.4		Benzene	lgK:2.8 (2SF)	3	MFP	148
2	6.2		Cyclohexane	lgK:3.7 (2SF)	3	MFP	148

Reaction No. 34365



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	6.2		Cyclohexane	lgK:-5.3 (2SF)	3	MFP	148

Reaction No. 34565



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:4.0 (0.04SD)w	5	MPH	3118
2	25	0	Inf. Dilution	lgK:4.1 (2SF)	1	ESC	

Reaction No. 37307							
Salicylaldoxime-2 + Ti+4 = Ti+4_Salicylaldoxime-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.5 (3SF)	4	MGL	931
2	25	0.01	KCl	lgK:18.29 (4SF)	2	MGL	931
3	25	0.025	KCl	lgK:17.74 (4SF)	4	MGL	931
4	25	0.05	KCl	lgK:17.35 (4SF)	4	MGL	931
5	25	0.075	KCl	lgK:16.86 (4SF)	0	MGL	931
6	25	0.1	KCl	lgK:16.30 (4SF)	0	MGL	931

Reaction No. 37308							
Ti+4_Salicylaldoxime-2 + Salicylaldoxime-2 = Ti+4_Salicylaldoxime-2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.2 (3SF)	3	MGL	931
2	25	0.01	KCl	lgK:16.88 (4SF)	3	MGL	931
3	25	0.025	KCl	lgK:16.62 (4SF)	2	MGL	931
4	25	0.05	KCl	lgK:16.07 (4SF)	2	MGL	931
5	25	0.075	KCl	lgK:15.66 (4SF)	0	MGL	931
6	25	0.1	KCl	lgK:14.85 (4SF)	0	MGL	931

Reaction No. 40404							
TiO+2_Oxalic-2 + Oxalic-2 = TiO+2_Oxalic-2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.02	HClO4	lgK:3.3 (2SF)	2	MSP	931
2	25	0	Inf. Dilution	lgK:3.3 (2SF)	1	EDH	
3	25	0.03	Unknown	lgK:5.11 (3SF)	0	MGL	931
4	25	0.03	Unknown	lgK:4.4 (2SF)	3	MSL	931

Reaction No. 41533							
Pb+2_TiO+2_Oxalic-2 (2) (s) = Pb+2 + TiO+2_Oxalic-2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NaClO4	lgK:-6.9 (0.03SD)	5	MSL	1872

Reaction No. 42110							
TiO+2 + H+1_Cat-2 = TiO+2_H+1_Cat-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	0.5	Unknown	lgK:4.143 (4SF)	5	MSP	4447

2	18:20	2	Unknown	lgK:4.252 (4SF)	5	MSP	4447
3	18:20	0.5	Ethanol Unknown	lgK:6.280 (4SF)	5	MSP	4447
4	18:20	1	Ethanol Unknown	lgK:6.086 (4SF)	5	MSP	4447
5	18:20	1	Ethanol:Water 25:75Vol Unknown	lgK:4.967 (4SF)	5	MSP	4447
6	18:20	2	Ethanol:Water 25:75Vol Unknown	lgK:4.222 (4SF)	5	MSP	4447
7	18:20	0.5	Ethanol:Water 50:50Vol Unknown	lgK:4.983 (4SF)	5	MSP	4447
8	18:20	1	Ethanol:Water 50:50Vol Unknown	lgK:5.076 (4SF)	5	MSP	4447
9	18:20	0.5	Ethanol:Water 75:25Vol Unknown	lgK:5.164 (4SF)	5	MSP	4447
10	18:20	1	Ethanol:Water 75:25Vol Unknown	lgK:5.046 (4SF)	5	MSP	4447
11	18:20	0.5	Methanol Unknown	lgK:6.041 (4SF)	5	MSP	4447
12	18:20	1	Methanol Unknown	lgK:6.107 (4SF)	5	MSP	4447
13	18:20	0.5	Methanol:Water 25:75Vol Unknown	lgK:4.124 (4SF)	5	MSP	4447
14	18:20	1	Methanol:Water 25:75Vol Unknown	lgK:4.648 (4SF)	5	MSP	4447
15	18:20	1	Methanol:Water 50:50Vol Unknown	lgK:5.158 (4SF)	5	MSP	4447
16	18:20	2	Methanol:Water 50:50Vol Unknown	lgK:5.036 (4SF)	5	MSP	4447
17	18:20	0.5	Methanol:Water 75:25Vol Unknown	lgK:5.644 (4SF)	5	MSP	4447
18	18:20	1	Methanol:Water 75:25Vol Unknown	lgK:5.509 (4SF)	5	MSP	4447
19	18:20	0.5	Propan-1-ol Unknown	lgK:6.415 (4SF)	5	MSP	4447
20	18:20	1	Propan-1-ol Unknown	lgK:6.355 (4SF)	5	MSP	4447

21	18:20	1	Propan-1-ol:Water 25:75Vol Unknown	lgK:4.292 (4SF)	5	MSP	4447
22	18:20	2	Propan-1-ol:Water 25:75Vol Unknown	lgK:4.425 (4SF)	5	MSP	4447
23	18:20	0.5	Propan-1-ol:Water 50:50Vol Unknown	lgK:4.476 (4SF)	5	MSP	4447
24	18:20	1	Propan-1-ol:Water 50:50Vol Unknown	lgK:4.625 (4SF)	5	MSP	4447
25	18:20	1	Propan-1-ol:Water 75:25Vol Unknown	lgK:5.041 (4SF)	5	MSP	4447
26	18:20	2	Propan-1-ol:Water 75:25Vol Unknown	lgK:5.089 (4SF)	5	MSP	4447
27	18:20	0.5	Propan-2-ol Unknown	lgK:6.717 (4SF)	5	MSP	4447
28	18:20	1	Propan-2-ol Unknown	lgK:6.780 (4SF)	5	MSP	4447
29	18:20	0.5	Propan-2-ol:Water 25:75Vol Unknown	lgK:4.244 (4SF)	5	MSP	4447
30	18:20	1	Propan-2-ol:Water 25:75Vol Unknown	lgK:4.409 (4SF)	5	MSP	4447
31	18:20	2	Propan-2-ol:Water 25:75Vol Unknown	lgK:4.208 (4SF)	5	MSP	4447
32	18:20	0.5	Propan-2-ol:Water 50:50Vol Unknown	lgK:4.260 (4SF)	5	MSP	4447
33	18:20	1	Propan-2-ol:Water 50:50Vol Unknown	lgK:4.409 (4SF)	5	MSP	4447
34	18:20	0.5	Propan-2-ol:Water 75:25Vol Unknown	lgK:4.602 (4SF)	5	MSP	4447
35	18:20	1	Propan-2-ol:Water 75:25Vol Unknown	lgK:4.742 (4SF)	5	MSP	4447

Reaction No. 42111

TiO+2 + H+1_Pyrogallol-3 = TiO+2_H+1_Pyrogallol-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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1	18:20	0.5	Ethanol Unknown	lgK:6.347 (4SF)	5	MSP	4447
2	18:20	1	Ethanol Unknown	lgK:6.416 (4SF)	5	MSP	4447
3	18:20	0.5	Ethanol:Water 25:75Vol Unknown	lgK:4.821 (4SF)	5	MSP	4447
4	18:20	1	Ethanol:Water 25:75Vol Unknown	lgK:4.688 (4SF)	5	MSP	4447
5	18:20	0.5	Ethanol:Water 50:50Vol Unknown	lgK:5.658 (4SF)	5	MSP	4447
6	18:20	1.5	Ethanol:Water 50:50Vol Unknown	lgK:5.721 (4SF)	5	MSP	4447
7	18:20	0.5	Ethanol:Water 75:25Vol Unknown	lgK:6.013 (4SF)	5	MSP	4447
8	18:20	1.5	Ethanol:Water 75:25Vol Unknown	lgK:6.061 (4SF)	5	MSP	4447
9	18:20	0.5	Methanol Unknown	lgK:6.240 (4SF)	5	MSP	4447
10	18:20	1	Methanol Unknown	lgK:6.155 (4SF)	5	MSP	4447
11	18:20	1.5	Methanol Unknown	lgK:6.119 (4SF)	5	MSP	4447
12	18:20	0.5	Methanol:Water 25:75Vol Unknown	lgK:5.058 (4SF)	5	MSP	4447
13	18:20	1	Methanol:Water 25:75Vol Unknown	lgK:5.173 (4SF)	5	MSP	4447
14	18:20	0.5	Methanol:Water 50:50Vol Unknown	lgK:5.352 (4SF)	5	MSP	4447
15	18:20	1	Methanol:Water 50:50Vol Unknown	lgK:5.292 (4SF)	5	MSP	4447
16	18:20	1.5	Methanol:Water 50:50Vol Unknown	lgK:5.611 (4SF)	5	MSP	4447
17	18:20	0.5	Methanol:Water 75:25Vol Unknown	lgK:5.668 (4SF)	5	MSP	4447
18	18:20	1.5	Methanol:Water 75:25Vol Unknown	lgK:5.876 (4SF)	5	MSP	4447

19	18:20	0.5	Propan-1-ol Unknown	lgK:6.456 (4SF)	5	MSP	4447
20	18:20	1	Propan-1-ol Unknown	lgK:6.509 (4SF)	5	MSP	4447
21	18:20	0.5	Propan-1-ol:Water 25:75Vol Unknown	lgK:4.450 (4SF)	5	MSP	4447
22	18:20	1	Propan-1-ol:Water 25:75Vol Unknown	lgK:4.577 (4SF)	5	MSP	4447
23	18:20	0.5	Propan-1-ol:Water 50:50Vol Unknown	lgK:4.658 (4SF)	5	MSP	4447
24	18:20	1	Propan-1-ol:Water 50:50Vol Unknown	lgK:4.757 (4SF)	5	MSP	4447
25	18:20	0.5	Propan-1-ol:Water 75:25Vol Unknown	lgK:5.556 (4SF)	5	MSP	4447
26	18:20	1.5	Propan-1-ol:Water 75:25Vol Unknown	lgK:5.719 (4SF)	5	MSP	4447
27	18:20	0.5	Propan-2-ol Unknown	lgK:6.721 (4SF)	5	MSP	4447
28	18:20	1	Propan-2-ol Unknown	lgK:4.462 (4SF)	5	MSP	4447
29	18:20	1	Propan-2-ol Unknown	lgK:6.796 (4SF)	5	MSP	4447
30	18:20	1.5	Propan-2-ol Unknown	lgK:4.553 (4SF)	5	MSP	4447
31	18:20	1	Propan-2-ol:Water 25:75Vol Unknown	lgK:4.699 (4SF)	5	MSP	4447
32	18:20	1.5	Propan-2-ol:Water 25:75Vol Unknown	lgK:4.688 (4SF)	5	MSP	4447
33	18:20	1	Propan-2-ol:Water 50:50Vol Unknown	lgK:4.699 (4SF)	5	MSP	4447
34	18:20	1.5	Propan-2-ol:Water 50:50Vol Unknown	lgK:4.785 (4SF)	5	MSP	4447
35	18:20	1	Propan-2-ol:Water 75:25Vol Unknown	lgK:5.516 (4SF)	5	MSP	4447
36	18:20	1.5	Propan-2-ol:Water 75:25Vol Unknown	lgK:5.512 (4SF)	5	MSP	4447

Reaction No. 43076							
Ti+4_OH-1(3) + 2<Lactic-1> = Ti+4_Lactic-1(2)_OH-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.3	NaCl	lgK:8.61(3SF)	4	MSP	906
2	25	0	Inf. Dilution	lgK:8.88(3SF)	2	EAE	

Reaction No. 43123							
Ala-1 + Ti+3 = Ti+3_Ala-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:8.50(3SF)w	3	MPH	906
2	23			lgK:8.53(3SF)	0	MGL	2392

Reaction No. 43131							
bAla-1 + Ti+3 = Ti+3_bAla-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:9.70(3SF)w	3	MPH	906
2	25	0	Inf. Dilution	lgK:13.8(3SF)	1	EAE	
3	25	1	NaClO4	lgK:13.28(4SF)	3	MEF	11516
Note (3): Yakovlev Y et al., Zhur. Neorg. Khim., 1977, 22, 87							

Reaction No. 43226							
Ti+4_OH-1 + Malic-2 = Ti+4_OH-1_Malic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:6.81(3SF)	4	MDS	906

Reaction No. 43227							
Ti+4_OH-1(3) + Malic-2 = Ti+4_OH-1(3)_Malic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.3	Unknown	lgK:6.74(0.02SD)	5	MSP	906
2	25	0	Inf. Dilution	lgK:7.28(3SF)	2	EAE	

Reaction No. 43267							
TiO+2 + H+1_Tartaric-2 = TiO+2_H+1_Tartaric-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	NaClO4	lgK:2.50(3SF)	4	MIX	906

Reaction No. 43268							
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TiO+2_H+1_Tartaric-2 + H+1_Tartaric-2 = TiO+2_H+1(2)_Tartaric-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	NaClO4	lgK:2.12(3SF)	4	MIX	906

Reaction No. 43420							
Asn-1 + Ti+3 = Ti+3_Asn-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:7.20(3SF)w	0	MPH	906
2	25	0.02	Unknown	lgK:6.95(3SF)	3	MGL	905

Reaction No. 43582							
Ti+4_OH-1 + 234TriOHPentDioic-2 = Ti+4_OH-1_234TriOHPentDioic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:5.45(3SF)	4	MSP	906

Reaction No. 43613							
Pro-1 + Ti+3 = Ti+3_Pro-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:10.05(4SF)w	3	MPH	906

Reaction No. 43641							
Val-1 + Ti+3 = Ti+3_Val-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:8.20(3SF)w	3	MPH	906
2	23			lgK:8.12(3SF)	0	MGL	2392

Reaction No. 43663							
2<Cat-2> + Ti+4 = Ti+4_Cat-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1.2	Unknown	lgK:50.1(3SF)	4	MSP	906

Reaction No. 43664							
Ti+4 + 2<Cat-2> + Oxalic-2 = Ti+4_Cat-2(2)_Oxalic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.5	HCl	lgK:59.4(3SF)	3	MSP	906
2	25	0	Inf. Dilution	lgK:59.7(3SF)	1	ESC	

Reaction No. 43665							
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Ti+4_OH-1(2) + Cat-2 + Oxalic-2 = Ti+4_OH-1(2)_Cat-2_Oxalic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:29.3(3SF)	0	MSP	906
2	25	0	Inf. Dilution	lgK:29.3(3SF)	1	ESC	

Reaction No. 43672							
Ti+4_Cat-2(3) + 2<H+1_Aniline> = Ti+4_H+1(2)_Aniline(2)_Cat-2(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:0.27(2SF)	4	MDS	906

Reaction No. 43748							
Ti+4_OH-1(2) + Citric-3 + H+1_Citric-3 = Ti+4_H+1_OH-1(2)_Citric-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.3	NaCl	lgK:16.3(0.05SD)	4	MSP	906
2	25	0	Inf. Dilution	lgK:17.4(3SF)	2	EAE	

Reaction No. 43749							
TiO+2 + H+1(2)_Citric-3 = TiO+2_H+1(2)_Citric-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	NaClO4	lgK:2.91(3SF)	4	MIX	906
2	25	0	Inf. Dilution	lgK:3.39(3SF)	2	EAE	

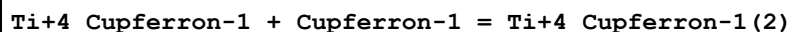
Reaction No. 43750							
TiO+2_H+1(2)_Citric-3 + H+1(2)_Citric-3 = TiO+2_H+1(4)_Citric-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	NaClO4	lgK:2.49(3SF)	4	MIX	906
2	25	0	Inf. Dilution	lgK:2.73(3SF)	2	EAE	

Reaction No. 43775							
Gluconic-1 + TiO+2 = TiO+2_Gluconic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22			lgK:4.43(3SF)	4	MSP	906
2	22			lgK:4.66(3SF)	4	MSP	906

Reaction No. 43982							
Cupferron-1 + Ti+4 = Ti+4_Cupferron-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

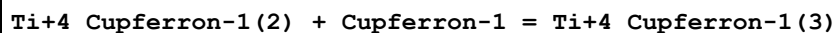
1	10	3	NaClO4	lgK:10.39(4SF)	4	MDS	906
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Reaction No. 43983



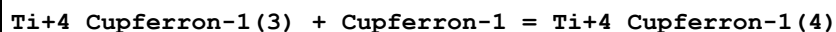
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	3	NaClO4	lgK:10.07(4SF)	4	MDS	906

Reaction No. 43984



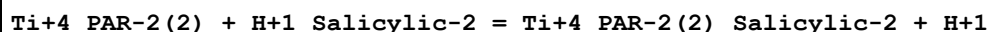
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	3	NaClO4	lgK:9.75(3SF)	4	MDS	906

Reaction No. 43985



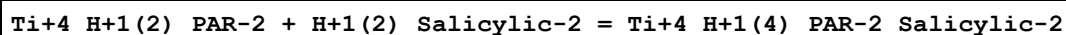
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	3	NaClO4	lgK:9.43(3SF)	4	MDS	906

Reaction No. 44197



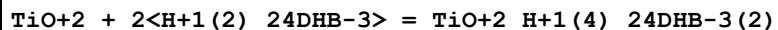
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	KCl	lgK:4.7(0.03SD)	5	MSP	906

Reaction No. 44198



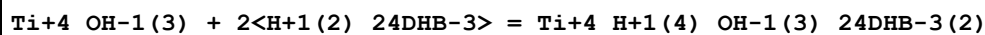
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	KCl	lgK:11.7(0.1SD)	5	MSP	906

Reaction No. 44213



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:6.0(0.03SD)	4	MSP	906

Reaction No. 44214



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	Unknown	lgK:6.5(0.03SD)	4	MSP	906

Reaction No. 44235

Ti+4_OH-1(3) + H+1(2)_234THB-4 = Ti+4_H+1(2)_OH-1(3)_234THB-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	17:19	0.1	Unknown	lgK:3.54(0.02SD)	4	MSP	906

Reaction No. 44236							
Ti+4_OH-1(3) + 2<H+1_234THB-4> = Ti+4_H+1(2)_OH-1(3)_234THB-4(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	17:19	0.1	Unknown	lgK:7.6(0.04SD)	4	MSP	906

Reaction No. 44342							
Ti+4_PAR-2(2) + H+1_SulfSal-3 = Ti+4_PAR-2(2)_SulfSal-3 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	KCl	lgK:3.44(0.01SD)	4	MSP	906
2	25	0	Inf. Dilution	lgK:3.5(2SF)	1	ESC	

Reaction No. 44343							
Ti+4_H+1(2)_PAR-2 + H+1(2)_SulfSal-3 = Ti+4_H+1(4)_PAR-2_SulfSal-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	KCl	lgK:11.4(0.1SD)	4	MSP	906
2	25	0	Inf. Dilution	lgK:11.5(3SF)	1	ESC	

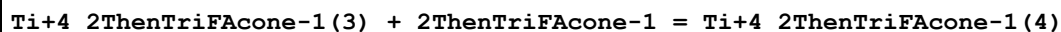
Reaction No. 44571							
2ThenTriFAcone-1 + Ti+4 = Ti+4_2ThenTriFAcone-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgK:7.87(3SF)	4	MDS	906
2	25	1	NaClO4	lgK:7.87(3SF)	4	MDS	906

Reaction No. 44572							
Ti+4_2ThenTriFAcone-1 + 2ThenTriFAcone-1 = Ti+4_2ThenTriFAcone-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgK:7.65(3SF)	4	MDS	906
2	25	1	NaClO4	lgK:7.65(3SF)	4	MDS	906

Reaction No. 44573							
Ti+4_2ThenTriFAcone-1(2) + 2ThenTriFAcone-1 = Ti+4_2ThenTriFAcone-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgK:7.45(3SF)	4	MDS	906

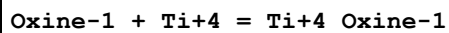
2	25	1	NaClO4	lgK:7.45 (3SF)	4	MDS	906
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Reaction No. 44574



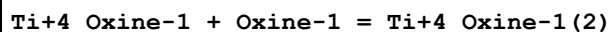
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgK:7.23 (3SF)	4	MDS	906
2	25	1	NaClO4	lgK:7.23 (3SF)	4	MDS	906

Reaction No. 44653



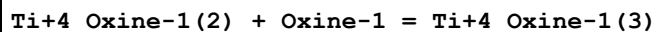
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HClO4	lgK:13.22 (4SF)	4	MDS	906
2	20	1	NaClO4	lgK:13.22 (4SF)	4	MDS	906

Reaction No. 44654



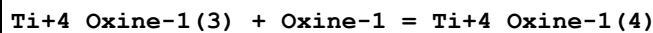
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HClO4	lgK:12.72 (4SF)	4	MDS	906
2	20	1	NaClO4	lgK:12.72 (4SF)	4	MDS	906

Reaction No. 44655



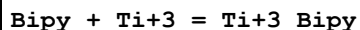
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HClO4	lgK:12.26 (4SF)	4	MDS	906
2	20	1	NaClO4	lgK:12.26 (4SF)	4	MDS	906

Reaction No. 44656



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HClO4	lgK:11.00 (4SF)	4	MDS	906
2	20	1	NaClO4	lgK:11.00 (4SF)	4	MDS	906

Reaction No. 44736



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	50	0.5	HCl	lgK:2.9 (0.04SD)	4	MSP	906
2	50	0.5	KCl	lgK:2.9 (0.04SD)	4	MSP	906

Reaction No. 44737							
2<Bipy> + Ti+3 = Ti+3_Bipy(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	50	0.5	HCl	lgK:4.8(0.05SD)	4	MSP	906
2	50	0.5	KCl	lgK:4.8(0.05SD)	4	MSP	906

Reaction No. 44738							
3<Bipy> + Ti+3 = Ti+3_Bipy(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	50	0.5	HCl	lgK:6.4(0.04SD)	4	MSP	906
2	50	0.5	KCl	lgK:6.4(0.04SD)	4	MSP	906

Reaction No. 44795							
Ti+4_OH-1(2) + 2<H+1(2)_Chromotropic-4> = Ti+4_H+1(2)_OH-1(2)_Chromotropic-4(2) + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:14.36(4SF)	4	MSP	906

Reaction No. 44796							
Ti+4_OH-1(3) + 3<H+1(2)_Chromotropic-4> = Ti+4_H+1(3)_OH-1_Chromotropic-4(3) + H+1 + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:25.92(4SF)	4	MSP	906

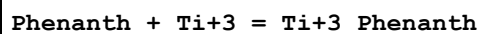
Reaction No. 44885							
Ti+3 + H+1(2)_EDTA-4 = Ti+3_H+1(2)_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1.2	KCl	lgK:7.5(0.09SD)	4	MSP	906
2	25	0	Inf. Dilution	lgK:9.23(3SF)	2	EAE	

Reaction No. 44886							
TiO+2 + H+1_EDTA-4 = TiO+2_H+1_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:22	1	NaClO4	lgK:12.08(0.02SD)	5	MSP	906
2	25	0	Inf. Dilution	lgK:13.9(3SF)	2	EAE	

Reaction No. 45063							
TiO+2_Acetic-1 + H+1_PAR-2 = TiO+2_H+1_Acetic-1_PAR-2							

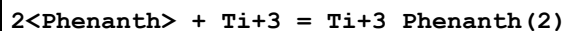
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:13.25 (4SF)	4	MSP	906

Reaction No. 45107



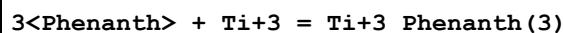
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20			lgK:3.78 (0.01SD)	4	MSP	906

Reaction No. 45108



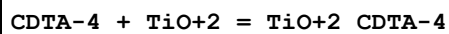
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20			lgK:8.29 (0.02SD)	4	MSP	906

Reaction No. 45109



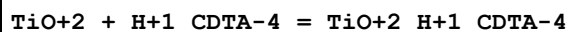
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20			lgK:12.50 (4SF)	4	MSP	906

Reaction No. 45254



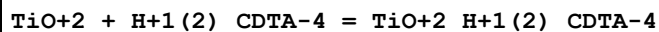
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:22	0.1	NaClO4	lgK:18.2 (0.1SD)	4	MSP	906
2	25	0	Inf. Dilution	lgK:18.6 (3SF)	1	ESC	

Reaction No. 45255



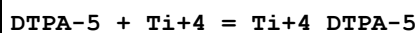
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:22	0.1	NaClO4	lgK:11.1 (0.1SD)	4	MSP	906
2	25	0	Inf. Dilution	lgK:11.4 (3SF)	1	ESC	

Reaction No. 45256



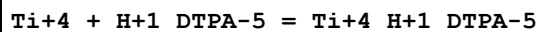
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:22	0.1	NaClO4	lgK:8.33 (0.02SD)	4	MSP	906
2	25	0	Inf. Dilution	lgK:8.5 (2SF)	1	ESC	

Reaction No. 45278



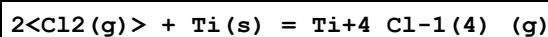
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:22			lgK:23.36(4SF)	0	MSP	906
2	18:22			lgK:23.38(4SF)	0	MSP	906
3	25	0	Inf. Dilution	lgK:23.3(3SF)	1	ESC	

Reaction No. 45279



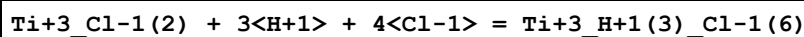
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:22			lgK:14.50(4SF)	0	MSP	906
2	18:22			lgK:14.51(4SF)	0	MSP	906
3	25	0	Inf. Dilution	lgK:14.5(3SF)	1	ESC	

Reaction No. 52228



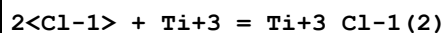
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-726.3(4SF) kJ	3	CRV	6330
2	25	0	Inf. Dilution	dG:-726.3(4SF) kJ	6	CRV	8746
3	25	0	Inf. Dilution	dG:-726.32(3.229SD) kJ	6	CRV	14923
4	25	0	Inf. Dilution	dH:-763.2(4SF) kJ	3	CRV	6330
5	25	0	Inf. Dilution	dH:-763.2(3.0MD) kJ	6	CRV	8746
6	25	0	Inf. Dilution	dH:-763.20(3.000SD) kJ	6	CRV	14923
7	25	0	Inf. Dilution	Cp:95.408(0.5SD) J/K	5	CRV	27292

Reaction No. 52990



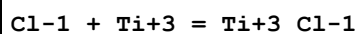
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:-7.8(2SF)	0	MSP	5398

Reaction No. 52991



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:0.37(2SF)	0	MSP	5398

Reaction No. 52992



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.925(3SF)	0	MSP	931

2	25	0	Unknown	lgK:0.1(1SF)	3	EOD	19339
Note (2): Range from 0.07 to 0.2							
3	25			lgK:0.56(2SF)	0	MSP	5398
4	40	0.5	Unknown	lgK:0.34(2SF)	5	CRV	2556

Reaction No. 52993							
Cl-1 + Ti+4 = Ti+4_Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	5	HClO4	lgK:0.60(2SF)	5	MSP	5398
2	25	6	HClO4	lgK:0.42(2SF)	5	MSP	5398
3	25	7	HClO4	lgK:0.0(1SF)	4	MSP	5398
4	25	7.5	HClO4	lgK:-0.05(1SF)	4	MSP	5398
5	25	8	HClO4	lgK:-0.23(2SF)	4	MSP	5398
6	20	3.5	KCl	lgR:-0.5(1SF)	5	MSL	5398

Reaction No. 53158							
Ti+4_F-1(5) + F-1 = Ti+4_F-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:3.79(3SF)	3	MIX	5398
2	25	0.5	HCl	lgK:3.8(0.1SD)	5	MEF	5398

Reaction No. 53159							
Ti+4_F-1(4)_OH-1 + H+1_F-1 + F-1 = Ti+4_F-1(6) + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0			lgK:6.48(3SF)	3	MKN	5398
2	25			lgK:5.85(3SF)	3	MKN	5398

Reaction No. 53300							
SO4-2 + TiO+2 = TiO+2_SO4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18			lgK:2.40(3SF)	0	MCE	140
2	20	8	H2SO4	lgK:2.23(3SF)	4	MDS	5398
3	25	0	Inf. Dilution	lgK:2.52(3SF)	4	MSP	5398
4	25	0	Unknown	lgK:2.4(0.05SD)	4	ENS	5558
5	25	3	HClO4	lgK:2.15(3SF)	5	MSP	5398
6	25	4	HClO4	lgK:2.26(3SF)	5	MSP	5398
7	25	5	HClO4	lgK:2.47(3SF)	5	MSP	5398

Reaction No. 53301							
$2\text{SO}_4\text{-2} + \text{TiO}_2 = \text{TiO}_2\text{SO}_4\text{-2(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	8	H2SO4	lgK:4.11 (3SF)	4	MDS	5398
2	25	4	HClO4	lgK:3.80 (3SF)	5	MSP	5398

Reaction No. 53501							
$\text{TiO}_2 + \text{HO}_2\text{-1} = \text{TiO}_2\text{HO}_2\text{-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18	0.3	Unknown	lgK:12.29 (4SF)	5	MKN	5398
2	25	0	Inf. Dilution	lgK:12.8 (3SF)	2	EAE	

Reaction No. 53502							
$\text{TiO}_2 + \text{H}_2\text{O}_2 = \text{TiO}_2\text{H}_2\text{O}_2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.71 (3SF)	4	MSP	5398
2	25	0	Inf. Dilution	dH:-10.5 (3SF)	4	MCL	5398

Reaction No. 53995							
$\text{O}_2\text{-2} + \text{Ti+4} = \text{Ti+4O}_2\text{-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	5	0	Inf. Dilution	lgK:4.29 (3SF)	3	MSP	931
2	5	1	HCl	lgK:4.42 (3SF)	5	MSP	931
3	15	0	Inf. Dilution	lgK:4.01 (3SF)	4	MSP	931
4	15	1	HCl	lgK:4.15 (3SF)	5	MSP	931
5	25	0	Inf. Dilution	lgK:3.74 (3SF)	4	MSP	931
6	25	0.5	HCl	lgK:3.78 (3SF)	5	MSP	931
7	25	0.5	HClO4	lgK:3.81 (3SF)	5	MSP	931
8	25	1	HCl	lgK:3.85 (3SF)	5	MSP	931
9	25	1	HCl	lgK:3.87 (3SF)	5	MSP	931
10	25	1	HClO4	lgK:3.87 (3SF)	5	MSP	931
11	25	2	Unknown	lgK:3.94 (3SF)	0	CRV	552
12	25	3	HCl	lgK:3.95 (3SF)	5	MSP	931
13	25	3	HClO4	lgK:4.14 (3SF)	5	MSP	931
14	25	3	Unknown	lgK:4.01 (3SF)	0	CRV	552
15	25	5	HCl	lgK:4.07 (3SF)	3	MSP	931

16	25	5	HClO ₄	lgK:4.46(3SF)	3	MSP	931
17	40	0	Inf. Dilution	lgK:3.44(3SF)	3	MSP	931
18	40	1	HCl	lgK:3.55(3SF)	5	MSP	931
19	5:40	0	Inf. Dilution	dH:-10.0(3SF)	4	MTD	931
20	25	0.1	Unknown	dH:-10.5(3SF)	6	CRV	552

Reaction No. 54116							
TiO ₂ _F-1(5) + F-1 = TiO ₂ _F-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.5	HCl	lgK:3.83(3SF)	0	MAX	931
Note (1): Rxn deprecated coz it is thermodynamically isolated							
2	25	0	Inf. Dilution	lgK:2.95(3SF)	0	EAE	

Reaction No. 54168							
Cl-1 + TiO ₂ = TiO ₂ _Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	3	NaClO ₄	lgK:0.55(2SF)	4	MCE	931
2	25	0	Unknown	lgK:3.55(0.35SD)	3	EOD	19339

Reaction No. 54169							
TiO ₂ _Cl-1 + Cl-1 = TiO ₂ _Cl-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	3	NaClO ₄	lgK:-0.40(2SF)	4	MCE	931
2	25	0	Unknown	lgK:0.40(0.06SD)	3	EOD	19339

Reaction No. 54170							
TiO ₂ _Cl-1(2) + Cl-1 = TiO ₂ _Cl-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:-1.03(3SF)	0	MCE	931
2	25	0	Unknown	lgK:10.6(0.20SD)	3	EOD	19339

Reaction No. 54171							
TiO ₂ _Cl-1(3) + Cl-1 = TiO ₂ _Cl-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:-1.1(2SF)	0	MCE	931
2	25	0	Unknown	lgK:12.(2SF)	3	EOD	19339

Reaction No. 54565							
$\text{Ti+4_OH-1(2)} + \text{F-1} = \text{Ti+4_F-1_OH-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	NaClO4	lgK:5.4(0.07SD)	5	CMN	437

Reaction No. 54566							
$\text{Ti+4_F-1_OH-1(2)} + \text{F-1} = \text{Ti+4_F-1(2)_OH-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	NaClO4	lgK:4.3(0.07SD)	5	CMN	437

Reaction No. 54567							
$\text{Ti+4_F-1(2)_OH-1(2)} + \text{F-1} = \text{Ti+4_F-1(3)_OH-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	NaClO4	lgK:4.0(0.07SD)	5	CMN	437

Reaction No. 54683							
$\text{Ti+4_OH-1(2)} + 2\text{<H+1>} + \text{e-1} = \text{Ti+3} + 2\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.5(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:1.7(0.05SD)	0	ENS	778
3	25	0	Inf. Dilution	E:0.0077(2SF)	0	ENS	7276

Reaction No. 54684							
$\text{Ti+4_H+1_OH-1(2)} + \text{H+1} + \text{e-1} = \text{Ti+3} + 2\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.9(0.05SD)	0	ENS	778

Reaction No. 55313							
$\text{O2(g)} + \text{Ti(s)} = \text{TiO2(Rutile,s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:155.8(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:155.7(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:154.8(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:154.7(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:113.7(4SF)	4	ENS	6422
6	127	0	Inf. Dilution	lgK:113.6(4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:89.03(4SF)	3	ENS	6422

8	227	0	Inf. Dilution	lgK:88.96(4SF)	3	ENS	6494
9	327	0	Inf. Dilution	lgK:72.60(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:72.56(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-212.6(0.05SD)	5	ENS	802
12	25	0	Inf. Dilution	dG:-889.41(5SF) kJ	4	EFE	5138
13	25	0	Inf. Dilution	dG:-212.6(4SF)	4	EFE	6422
14	25	0	Inf. Dilution	dG:-212.59(5SF)	5	EFE	6482
15	25	0	Inf. Dilution	dG:-212.4(4SF)	4	EFE	6494
16	25	0	Inf. Dilution	dG:-889.5(4SF) kJ	4	EFE	7381
17	25	0	Inf. Dilution	dG:-212.6(4SF)	5	CRV	7382
18	25	0	Inf. Dilution	dG:-888.8(4SF) kJ	6	CRV	8746
19	25	0	Inf. Dilution	dG:-888.77(0.806SD) kJ	6	CRV	14923
20	25	0	Inf. Dilution	dG:-889.5(4SF) kJ	4	EOD	16743
21	25	0	Inf. Dilution	dG:-889.5(4SF) kJ	3	ENS	19339
22	25	0	Inf. Dilution	dG:-212.47(5SF)	3	CRV	31291
23	25	0	Inf. Dilution	dH:-225.8(0.1SD)	5	ENS	802
24	25	0	Inf. Dilution	dH:-944.75(1.3SD) kJ	4	MTD	5138
25	25	0	Inf. Dilution	dH:-225.8(4SF)	4	MTD	6422
26	25	0	Inf. Dilution	dH:-225.8(4SF)	4	ENS	6437
27	25	0	Inf. Dilution	dH:-255.8(4SF)	0	EFE	6482
Note (27): Low wgt for internal consistency							
28	25	0	Inf. Dilution	dH:-225.6(4SF)	4	MTD	6494
29	25	0	Inf. Dilution	dH:-944.7(1.3SD) kJ	4	ENS	7381
30	25	0	Inf. Dilution	dH:-944.0(0.8MD) kJ	6	CRV	8746
31	25	0	Inf. Dilution	dH:-944.00(0.800SD) kJ	6	CRV	14923
32	25	0	Inf. Dilution	dH:-944.7(4SF) kJ	4	CRV	16743
33	127	0	Inf. Dilution	dH:-225.7(4SF)	4	MTD	6422
34	127	0	Inf. Dilution	dH:-225.5(4SF)	4	MTD	6494
35	227	0	Inf. Dilution	dH:-225.5(4SF)	3	MTD	6422
36	227	0	Inf. Dilution	dH:-225.3(4SF)	3	MTD	6494
37	327	0	Inf. Dilution	dH:-225.3(4SF)	0	MTD	6422
38	327	0	Inf. Dilution	dH:-225.1(4SF)	0	MTD	6494
39	25	0	Inf. Dilution	Cp:55.10(0.1SD) J/K	4	ENS	5138

Reaction No. 55314

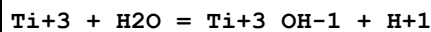
Ti+4 + 4<OH-1> = Ti+4_OH-1(4)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	3.5	NaClO ₄	lgK:58.3(0.05SD)	5	MSL	5499
2	20	3.5	Unknown	lgK:58.3(3SF)	3	EOD	20328
3	25	0	Inf. Dilution	lgK:53.10(4SF)	3	ENS	773

Reaction No. 55315							
O ₂ (g) + Ti(s) = TiO ₂ (Anatase,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:145.6(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:154.7(4SF)	0	ENS	6422
3	25	0	Inf. Dilution	lgK:154.7(4SF)	0	ENS	6494
4	27	0	Inf. Dilution	lgK:153.7(4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:153.7(4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:112.9(4SF)	0	ENS	6422
7	127	0	Inf. Dilution	lgK:112.9(4SF)	0	ENS	6494
8	227	0	Inf. Dilution	lgK:88.38(4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:88.38(4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:72.07(4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:72.07(4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-211.4(0.05SD)	0	ENS	802
13	25	0	Inf. Dilution	dG:-211.4(0.05SD)	0	ENS	802
14	25	0	Inf. Dilution	dG:-211.1(4SF)	0	EFE	6422
15	25	0	Inf. Dilution	dG:-211.1(4SF)	0	EFE	6494
16	25	0	Inf. Dilution	dG:-884.5(4SF) kJ	0	EFE	7381
17	25	0	Inf. Dilution	dG:-211.4(4SF)	0	CRV	7382
18	25	0	Inf. Dilution	dG:-884.5(4SF) kJ	0	ENS	19339
19	25	0	Inf. Dilution	dG:-211.36(5SF)	0	CRV	31291
20	25	0	Inf. Dilution	dH:-224.6(0.1SD)	0	ENS	802
21	25	0	Inf. Dilution	dH:-224.6(0.1SD)	0	ENS	802
22	25	0	Inf. Dilution	dH:-224.4(4SF)	2	MTD	6422
23	25	0	Inf. Dilution	dH:-224.4(4SF)	0	MTD	6494
24	25	0	Inf. Dilution	dH:-939.7(4SF) kJ	0	ENS	7381
25	127	0	Inf. Dilution	dH:-224.3(4SF)	0	MTD	6422
26	127	0	Inf. Dilution	dH:-224.2(4SF)	0	MTD	6494
27	227	0	Inf. Dilution	dH:-224.1(4SF)	0	MTD	6422
28	227	0	Inf. Dilution	dH:-224.0(4SF)	0	MTD	6494

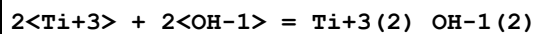
29	327	0	Inf. Dilution	dH:-223.8 (4SF)	0	MTD	6422
30	327	0	Inf. Dilution	dH:-223.8 (4SF)	0	MTD	6494

Reaction No. 55317



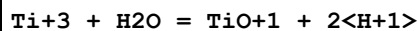
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.2 (2SF)	3	CRV	6478
2	25	0	Inf. Dilution	lgK:-1.29 (3SF)	2	CRV	32224
3	25	0	ClO4-1 salt	lgK:-1.3 (0.05SD)w	2	MPH	5498
4	25	0.5	ClO4-1 salt	lgK:-1.9 (0.05SD)w	2	MPH	5498
5	25	1	ClO4-1 salt	lgK:-2.5 (0.1SD)	2	MKN	5498
6	25	3	ClO4-1 salt	lgK:-2.8 (0.05SD)w	2	MPH	5498
7	25	0	Inf. Dilution	dH:10.1 (3SF)	3	CRV	6478

Reaction No. 55318



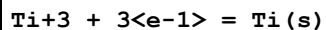
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Unknown	lgK:24.8 (0.1SD)	5	ENS	8
2	25	1	Unknown	lgK:24.6 (3SF)	5	CRV	2556
3	25	3	KCl	lgK:24.7 (0.03SD)	5	CRV	2556
4	25	3	Unknown	lgK:25.3 (0.05SD)	4	CSM	233

Reaction No. 55319



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.5 (0.05SD)	4	MSP	5398
2	25	0	Inf. Dilution	lgK:-4.5 (2SF)	4	CRV	18148

Reaction No. 55320



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-69.5 (0.05SD)	0	ENS	802
2	25	0	Inf. Dilution	lgK:-54.20 (4SF)	0	CRV	32224
3	25	0	Inf. Dilution	dG:350. (3SF)kJ	5	CRV	7421
4	25	0	Inf. Dilution	dG:350. (3SF)kJ	3	EFE	19339
5	25	0	Inf. Dilution	dG:415. (3SF)kJ	0	EOD	20358
6	25	0	Inf. Dilution	E:-1.37 (3SF)	0	CRV	6330

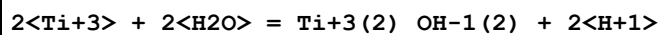
7	25	0	Inf. Dilution	E:-1.37(3SF)	0	CRV	22519
8	25	0	Inf. Dilution	E:-1.209(4SF)	3	CRV	22551
9	25	0	Inf. Dilution	dH:453.(3SF)kJ	3	EOD	20358

Reaction No. 55321



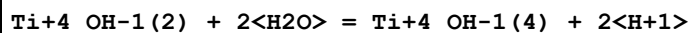
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.8(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-0.9(0.05SD)	0	ENS	802
3	25	0	Inf. Dilution	E:-0.055(2SF)	0	CRV	6330
4	25	0	Inf. Dilution	E:-0.055(2SF)	0	CRV	22519

Reaction No. 55325



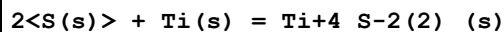
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.6(2SF)	4	CRV	32224
2	25	3	KCl	lgK:-3.9(0.05SD)	0	MGL	5398

Reaction No. 55328



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.6(1SF)	1	EES	
2	25	0.1	Unknown	lgK:-4.8(0.05SD)	0	ENS	4998
3	25	3	Unknown	lgK:1.3(2SF)	1	EES	

Reaction No. 55787



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:70.46(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:70.02(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:52.28(4SF)	4	ENS	6422
4	147	0	Inf. Dilution	lgK:49.72(4SF)	4	ENS	6422
5	25	0	Inf. Dilution	dG:-96.12(4SF)	4	EFE	6422
6	25	0	Inf. Dilution	dG:-430.3(4SF)kJ	0	EFE	19339
7	25	0	Inf. Dilution	dH:-97.3(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-98.3(3SF)	4	MTD	6422
9	147	0	Inf. Dilution	dH:-98.4(3SF)	4	MTD	6422

Reaction No. 55788							
S(s) + Ti(s) = Ti+2_S-2_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:47.32 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:47.02 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:35.17 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:27.99 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:23.17 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-64.55 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-65.0 (3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-65.6 (3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-66.0 (3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-66.3 (3SF)	4	MTD	6422

Reaction No. 56353							
Ba(s) + Ti(s) + 1.5<O2(g)> = BaTiO3(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:282.6 (4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:275.5 (4SF)	0	ENS	6422
3	27	0	Inf. Dilution	lgK:273.7 (4SF)	0	ENS	6422
4	121.6	0	Inf. Dilution	lgK:204.4 (4SF)	0	ENS	6422
5	25	0	Inf. Dilution	dG:-375.8 (0.05SD)	0	ENS	802
6	25	0	Inf. Dilution	dG:-375.8 (4SF)	0	EFE	6422
7	25	0	Inf. Dilution	dH:-396.7 (0.1SD)	0	ENS	802
8	25	0	Inf. Dilution	dH:-396.7 (4SF)	0	MTD	6422
9	121.6	0	Inf. Dilution	dH:-396.5 (4SF)	0	MTD	6422

Reaction No. 56376							
2<Mg(s)> + Ti(s) + 2<O2(g)> = Mg2TiO4(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:358.8 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:356.4 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:262.2 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:205.7 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:168.0 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-489.2 (0.05SD)	4	ENS	802

7	25	0	Inf. Dilution	dG:-489.2 (0.05SD)	5	ENS	802
8	25	0	Inf. Dilution	dG:-489.4 (4SF)	4	EFE	6422
9	25	0	Inf. Dilution	dH:-517.5 (0.1SD)	4	ENS	802
10	25	0	Inf. Dilution	dH:-517.5 (0.1SD)	5	ENS	802
11	25	0	Inf. Dilution	dH:-517.3 (4SF)	4	MTD	6422
12	127	0	Inf. Dilution	dH:-517.2 (4SF)	4	MTD	6422
13	227	0	Inf. Dilution	dH:-517.0 (4SF)	4	MTD	6422
14	327	0	Inf. Dilution	dH:-516.7 (4SF)	4	MTD	6422

Reaction No. 56377							
Mg(s) + Ti(s) + 1.5<O2(g)> = MgTiO3(geik,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:260.0 (4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:260.0 (4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:258.3 (4SF)	4	ENS	6422
4	27	0	Inf. Dilution	lgK:258.4 (4SF)	4	ENS	6494
5	127	0	Inf. Dilution	lgK:189.9 (4SF)	4	ENS	6422
6	127	0	Inf. Dilution	lgK:189.9 (4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:148.8 (4SF)	4	ENS	6422
8	227	0	Inf. Dilution	lgK:148.9 (4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:121.5 (4SF)	4	ENS	6422
10	327	0	Inf. Dilution	lgK:121.5 (4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-354.8 (0.05SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-354.8 (0.05SD)	5	ENS	802
13	25	0	Inf. Dilution	dG:-354.7 (4SF)	4	EFE	6422
14	25	0	Inf. Dilution	dG:-354.8 (4SF)	4	EFE	6494
15	25	0	Inf. Dilution	dH:-375.9 (0.1SD)	4	ENS	802
16	25	0	Inf. Dilution	dH:-375.9 (0.1SD)	5	ENS	802
17	25	0	Inf. Dilution	dH:-375.8 (4SF)	4	MTD	6422
18	25	0	Inf. Dilution	dH:-375.9 (4SF)	4	MTD	6494
19	127	0	Inf. Dilution	dH:-375.8 (4SF)	4	MTD	6422
20	127	0	Inf. Dilution	dH:-375.8 (4SF)	4	MTD	6494
21	227	0	Inf. Dilution	dH:-375.5 (4SF)	4	MTD	6422
22	227	0	Inf. Dilution	dH:-375.5 (4SF)	4	MTD	6494
23	327	0	Inf. Dilution	dH:-375.2 (4SF)	4	MTD	6422
24	327	0	Inf. Dilution	dH:-375.2 (4SF)	4	MTD	6494

Reaction No. 56393							
$\text{Sr(s)} + \text{Ti(s)} + 1.5\text{O}_2(\text{g}) = \text{SrTiO}_3(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:278.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:276.3(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:203.5(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:159.9(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:130.8(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-379.6(0.05SD)	4	ENS	802
7	25	0	Inf. Dilution	dG:-379.6(0.05SD)	5	ENS	802
8	25	0	Inf. Dilution	dG:-379.4(4SF)	4	EFE	6422
9	25	0	Inf. Dilution	dH:-399.7(0.1SD)	4	ENS	802
10	25	0	Inf. Dilution	dH:-399.7(0.1SD)	5	ENS	802
11	25	0	Inf. Dilution	dH:-399.7(4SF)	4	MTD	6422
12	127	0	Inf. Dilution	dH:-399.5(4SF)	4	MTD	6422
13	227	0	Inf. Dilution	dH:-399.2(4SF)	4	MTD	6422
14	327	0	Inf. Dilution	dH:-398.9(4SF)	4	MTD	6422

Reaction No. 56394							
$2\text{Sr(s)} + \text{Ti(s)} + 2\text{O}_2(\text{g}) = \text{Sr}_2\text{TiO}_4(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:380.2(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:377.7(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:278.2(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:218.5(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:178.7(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-520.7(0.05SD)	4	ENS	802
7	25	0	Inf. Dilution	dG:-520.7(0.05SD)	5	ENS	802
8	25	0	Inf. Dilution	dG:-518.7(4SF)	4	EFE	6422
9	25	0	Inf. Dilution	dH:-546.7(0.1SD)	4	ENS	802
10	25	0	Inf. Dilution	dH:-546.7(0.1SD)	5	ENS	802
11	25	0	Inf. Dilution	dH:-546.7(4SF)	4	MTD	6422
12	127	0	Inf. Dilution	dH:-546.4(4SF)	4	MTD	6422
13	227	0	Inf. Dilution	dH:-545.9(4SF)	4	MTD	6422
14	327	0	Inf. Dilution	dH:-545.4(4SF)	4	MTD	6422

Reaction No. 56403							
O2(g) + Ti(s) = TiO2(aq.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-196.3(0.05SD)	5	ENS	3006
2	25	0	Inf. Dilution	dG:-821.3(4SF)kJ	3	ENS	19339
3	25	0	Inf. Dilution	dH:-207.0(0.1SD)	5	ENS	3006

Reaction No. 56404							
1.5<O2(g)> + 2<Ti(s)> = Ti2O3(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:251.2(4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:249.6(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:183.4(4SF)	2	ENS	6422
4	197	0	Inf. Dilution	lgK:153.8(4SF)	2	ENS	6422
5	25	0	Inf. Dilution	dG:-342.3(0.1SD)	3	ENS	3006
6	25	0	Inf. Dilution	dG:-342.7(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-342.8(4SF)	3	CRV	7382
8	25	0	Inf. Dilution	dG:-1432.2(5SF)kJ	3	ENS	19339
9	25	0	Inf. Dilution	dG:-1434.2(5SF)kJ	3	ENS	19339
10	25	0	Inf. Dilution	dG:-342.73(5SF)	3	CRV	31291
11	25	0	Inf. Dilution	dH:-367.0(0.1SD)	3	ENS	3006
12	25	0	Inf. Dilution	dH:-363.5(4SF)	0	MTD	6422
13	127	0	Inf. Dilution	dH:-363.2(4SF)	1	MTD	6422
14	197	0	Inf. Dilution	dH:-362.5(4SF)	1	MTD	6422

Reaction No. 56405							
2.5<O2(g)> + 3<Ti(s)> = Ti3O5(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:406.0(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:403.3(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:296.3(4SF)	3	ENS	6422
4	177	0	Inf. Dilution	lgK:260.7(4SF)	3	ENS	6422
5	25	0	Inf. Dilution	dG:-553.1(0.05SD)	3	ENS	3006
6	25	0	Inf. Dilution	dG:-553.1(0.05SD)	4	ENS	3006
7	25	0	Inf. Dilution	dG:-553.9(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dG:-2314.(4SF)kJ	3	ENS	19339

9	25	0	Inf. Dilution	dG:-2317.4 (5SF) kJ	3	ENS	19339
10	25	0	Inf. Dilution	dG:-553.94 (5SF)	3	CRV	31291
11	25	0	Inf. Dilution	dH:-583.9 (0.1SD)	3	ENS	3006
12	25	0	Inf. Dilution	dH:-587.8 (4SF)	0	MTD	6422
13	127	0	Inf. Dilution	dH:-587.3 (4SF)	3	MTD	6422
14	177	0	Inf. Dilution	dH:-586.9 (4SF)	3	MTD	6422

Reaction No. 56406							
$\text{Ti(s)} + 1.5\text{O}_2(\text{g}) + 1.5\text{H}_2(\text{g}) = \text{Ti}_3\text{OH-1(3)}(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-250.9 (0.05SD)	5	ENS	3006
2	25	0	Inf. Dilution	dG:-1049.8 (5SF) kJ	3	ENS	19339

Reaction No. 56407							
$\text{F}_2(\text{g}) + \text{Ti(s)} = \text{Ti}_2\text{F-1(2)}(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-187.1 (0.05SD)	5	ENS	3006
2	25	0	Inf. Dilution	dG:-782.8 (4SF) kJ	3	ENS	19339
3	25	0	Inf. Dilution	dH:-198.0 (0.1SD)	5	ENS	3006

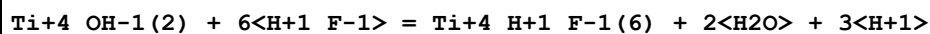
Reaction No. 57152							
$2\text{H}_2(\text{g}) + 2\text{O}_2(\text{g}) + \text{Ti(s)} = \text{Ti}_4\text{OH-1(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:224.4 (4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1309.10 (0.1SD) kJ	0	ENS	5138
3	25	0	Inf. Dilution	dG:-1312.5 (5SF) kJ	0	EOD	12843
4	25	0	Inf. Dilution	dG:-1312.50 (6SF) kJ	0	MSL	16743
5	25	0	Inf. Dilution	dH:-1511.80 (6SF) kJ	2	MTD	12843
6	25	0	Inf. Dilution	dH:-1515.30 (6SF) kJ	2	MSL	16743

Reaction No. 57162							
$\text{Ti}_4\text{OH-1(2)} + \text{H}_2\text{F-1} = \text{Ti}_4\text{F-1OH-1(2)} + \text{H}_2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	NaCl	lgK:2.3 (0.03SD)	6	MIS	5113

Reaction No. 57163							
$\text{Ti}_4\text{OH-1(2)} + 4\text{H}_2\text{F-1} = \text{Ti}_4\text{F-1(4)} + 2\text{H}_2\text{O} + 2\text{H}_2$							

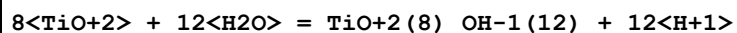
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	NaCl	lgK:8.34(0.02SD)	6	MIS	5113

Reaction No. 57164



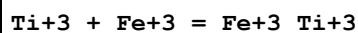
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	NaCl	lgK:11.41(0.01SD)	6	MIS	5113

Reaction No. 57431



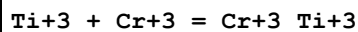
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	(H,Na)Cl	lgK:-1.68(0.1SD)	5	MEF	5272

Reaction No. 57530



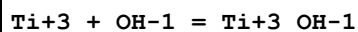
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19	0.04	H2SO4	lgK:1.4(0.2SD)	4	MSP	5309
2	25	0	Inf. Dilution	lgK:0.0413(3SF)	2	EAE	

Reaction No. 57531



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	26	0.03	H2SO4	lgK:1.5(0.2SD)	4	MSP	5309

Reaction No. 57882



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18	1	CsCl	lgK:12.30(4SF)	0	MSL	20361
2	25	0	Inf. Dilution	lgK:12.5(3SF)	1	EOR	
3	25	0	Inf. Dilution	lgK:11.5(0.1SD)	0	ENS	8
4	25	0	Inf. Dilution	lgK:12.7(3SF)	0	CRV	552
5	25	0.5	Unknown	lgK:11.8(0.1SD)	0	CRV	2556
6	25	1	Unknown	lgK:11.4(0.2SD)	0	CRV	2556
7	25	3	KCl	lgK:11.5(0.1SD)	3	CRV	2556
8	10:40	0	Inf. Dilution	dH:-14.(2SF)	2	CRV	2556

Reaction No. 57883

TiO ₂ + H ₂ O = TiO ₂ _OH-1 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.82 (4SF)	0	CRV	18148
Note (1): Wrong reaction. lgK refers to Kw/Krxn							
2	25	0	Inf. Dilution	lgK:-1.30 (3SF)	5	EOD	20328

Reaction No. 57884							
Ti+4_OH-1 (3) + H+1 = Ti+4_OH-1 (2) + H ₂ O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.15 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:0.3 (1SF)	0	CRV	18148
Note (2): Surely wrong conditions							
3	25	0	Inf. Dilution	lgK:1.71 (3SF)	0	MSL	31297
4	25	0	Unknown	lgK:0.3 (0.1SD)	0	ENS	5581
5	25	1.5	HClO ₄	lgR:0.30 (2SF)	0	MIX	5557
Note (5): Temp unspecified							
6	25	0	Inf. Dilution	dH:-18.0 (3SF) kJ	2	MTD	31297

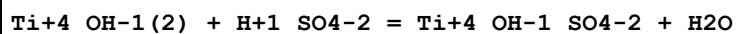
Reaction No. 57885							
TiO ₂ + H+1_SO4-2 = TiO ₂ _H+1_SO4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.5 (2SF)	4	CRV	7271
2	25	0	Unknown	lgK:2.2 (0.05SD)	0	CSM	8
3	25	3	HClO ₄	lgK:2.15 (3SF)	5	CRV	7271
4	25	4	HClO ₄	lgK:2.26 (3SF)	5	CRV	7271
5	25	5	HClO ₄	lgK:2.47 (3SF)	5	CRV	7271

Reaction No. 57886							
Ti+3 + SO4-2 = Ti+3_SO4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18	3	Unknown	lgK:1.4 (0.1SD)	3	MIX	5225
2	25	0	Unknown	lgK:1.0 (0.05SD)	3	ENS	5580
3	25	1.12	Unknown	lgK:1.4 (0.05SD)	2	MSP	5225

Reaction No. 57887							
Ti+4_OH-1 (3) + H+1_SO4-2 = Ti+4_OH-1 (2)_SO4-2 + H ₂ O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

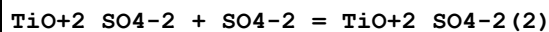
1	23	2	H2SO4	lgK:1.05 (3SF)	4	MIX	5557
2	25	0	Inf. Dilution	lgK:1.29 (3SF)	2	EAE	

Reaction No. 57888



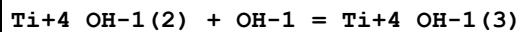
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	H2SO4	lgK:-0.20 (2SF)	4	MIX	5557
2	25	0	Inf. Dilution	lgK:0.284 (3SF)	2	EAE	

Reaction No. 57889



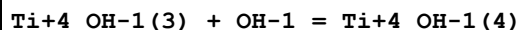
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.2 (2SF)	1	ELC	
2	25	0	Unknown	lgK:2.2 (0.05SD)	0	ENS	5558

Reaction No. 57890



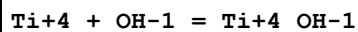
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.0 (3SF)	1	EES	
2	25	0	Inf. Dilution	lgK:11.51 (4SF)	0	CRV	20328
3	25	0.1	KNO3	lgK:13.4 (0.05SD)	4	MPS	5398
4	25	0.3	KNO3	lgK:13.6 (0.05SD)	4	MPS	5398
5	25	0.5	KNO3	lgK:13.8 (0.05SD)	4	MPS	5398
6	25	1	KNO3	lgK:14.0 (0.05SD)	4	MPS	5398

Reaction No. 57891



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18	0.1	Unknown	lgK:10.77 (4SF)	0	EOD	20328
2	25	0.1	KNO3	lgK:13.1 (0.05SD)	5	MPS	5398
3	25	0.3	KNO3	lgK:13.3 (0.05SD)	5	MPS	5398
4	25	0.5	KNO3	lgK:13.5 (0.05SD)	5	MPS	5398
5	25	1	KNO3	lgK:13.7 (0.05SD)	5	MPS	5398

Reaction No. 57892



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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1	18	0.1	Unknown	lgK:17.96(4SF)	0	EOD	20328
2	25	0	Inf. Dilution	lgK:17.96(4SF)	0	CRV	20328
3	25	0.1	KNO3	lgK:14.2(0.05SD)	4	MPS	5398
4	25	0.3	KNO3	lgK:14.3(0.05SD)	4	MPS	5398
5	25	0.5	KNO3	lgK:14.4(0.05SD)	4	MPS	5398
6	25	1	KNO3	lgK:14.7(0.05SD)	4	MPS	5398

Reaction No. 57893							
Ti+4_OH-1 + OH-1 = Ti+4_OH-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.6(3SF)	0	EES	
2	25	0	Inf. Dilution	lgK:17.21(4SF)	0	CRV	20328
3	25	0.1	KNO3	lgK:13.7(0.05SD)	3	MPS	5398
4	25	0.3	KNO3	lgK:13.9(0.05SD)	4	MPS	5398
5	25	0.5	KNO3	lgK:14.0(0.05SD)	4	MPS	5398
6	25	1	KNO3	lgK:14.3(0.05SD)	4	MPS	5398

Reaction No. 57894							
Ti2O5+2 + H2O = Ti2O5+2_OH-1 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:-2.5(0.05SD)	5	MGL	20374
Note (1): Ti2O5+2 is formed by peroxide; best model like this with H2O2 implicit							

Reaction No. 57895							
Ti2O5+2 + 2<H2O> = Ti2O5+2_OH-1(2) + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:-4.6(0.05SD)	5	MGL	20374

Reaction No. 57896							
Ti2O5+2 + 3<H2O> = Ti2O5+2_OH-1(3) + 3<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:-8.7(0.1SD)	5	MGL	20374

Reaction No. 57897							
Ti2O5+2 + 4<H2O> = Ti2O5+2_OH-1(4) + 4<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:-14.3(0.1SD)	5	MGL	20374

Reaction No. 57898							
$\text{Ti}_2\text{O}_5 + 2 + 5\text{H}_2\text{O} = \text{Ti}_2\text{O}_5 + 2\text{OH}^{-1}(5) + 5\text{H}^{+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK: -21.5 (0.1SD)	5	MGL	20374

Reaction No. 57899							
$\text{Ti}_2\text{O}_5 + 2 + 6\text{H}_2\text{O} = \text{Ti}_2\text{O}_5 + 2\text{OH}^{-1}(6) + 6\text{H}^{+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK: -30.0 (0.1SD)	5	MGL	20374

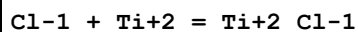
Reaction No. 57900							
$0.5\text{O}_2(\text{g}) + \text{Ti}(\text{s}) - 2\text{e}^{-1} = \text{TiO} + 2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK: 101.2 (4SF)	2	EOR	
2	25	0	Inf. Dilution	dG: -138.0 (0.1SD)	2	ENS	3006
3	25	0	Inf. Dilution	dG: -577.4 (4SF) kJ	2	EFE	19339
4	25	0	Inf. Dilution	dH: -689.9 (4SF) kJ	2	EOD	20358

Reaction No. 57901							
$\text{Ti} + 3 + 2\text{SO}_4^{-2} = \text{Ti} + 3\text{SO}_4^{-2}(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18	3	Unknown	lgK: 2.3 (0.1SD)	6	MIX	5225
2	25	0	Inf. Dilution	lgK: 2.5 (2SF)	1	ELC	

Reaction No. 57911							
$\text{Fe}(\text{s}) + 1.5\text{O}_2(\text{g}) + \text{Ti}(\text{s}) = \text{FeTiO}_3(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK: 202.9 (4SF)	3	ENS	6422
2	25	0	Inf. Dilution	lgK: 202.4 (4SF)	3	ENS	6494
3	27	0	Inf. Dilution	lgK: 201.6 (4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK: 201.1 (4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK: 147.8 (4SF)	3	ENS	6422
6	127	0	Inf. Dilution	lgK: 147.5 (4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK: 115.6 (4SF)	3	ENS	6422
8	227	0	Inf. Dilution	lgK: 115.3 (4SF)	3	ENS	6494
9	327	0	Inf. Dilution	lgK: 94.10 (4SF)	0	ENS	6422

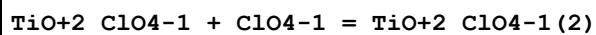
10	327	0	Inf. Dilution	lgK:93.95 (4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-278. (1.SD)	0	ENS	3006
12	25	0	Inf. Dilution	dG:-276.8 (4SF)	3	EFE	6422
13	25	0	Inf. Dilution	dG:-276.13 (5SF)	3	EFE	6482
14	25	0	Inf. Dilution	dG:-276.1 (4SF)	3	EFE	6494
15	25	0	Inf. Dilution	dG:-1155.5 (5SF) kJ	3	EOD	12843
16	25	0	Inf. Dilution	dG:-1168.8 (5SF) kJ	0	ENS	19339
17	25	0	Inf. Dilution	dG:-1158.1 (5SF) kJ	3	ENS	19339
18	25	0	Inf. Dilution	dG:-1267. (4SF) kJ	0	ENS	19339
19	25	0	Inf. Dilution	dG:-1278. (4SF) kJ	0	ENS	19339
20	25	0	Inf. Dilution	dG:-1125.1 (5SF) kJ	0	ENS	19339
21	25	0	Inf. Dilution	dH:-295. (1.SD)	4	ENS	3006
22	25	0	Inf. Dilution	dH:-295. (1.SD)	5	ENS	3006
23	25	0	Inf. Dilution	dH:-295.3 (4SF)	3	MTD	6422
24	25	0	Inf. Dilution	dH:-294.44 (5SF)	4	EFE	6482
25	25	0	Inf. Dilution	dH:-294.5 (4SF)	4	MTD	6494
26	25	0	Inf. Dilution	dH:-1232.0 (5SF) kJ	3	EOD	12843
27	127	0	Inf. Dilution	dH:-295.1 (4SF)	4	MTD	6422
28	127	0	Inf. Dilution	dH:-294.2 (4SF)	4	MTD	6494
29	227	0	Inf. Dilution	dH:-294.7 (4SF)	3	MTD	6422
30	227	0	Inf. Dilution	dH:-293.9 (4SF)	3	MTD	6494
31	327	0	Inf. Dilution	dH:-294.4 (4SF)	0	MTD	6422
32	327	0	Inf. Dilution	dH:-293.5 (4SF)	0	MTD	6494

Reaction No. 58573



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.955 (3SF)	2	EAE	
2	40	0.5	HClO4	lgK:0.34 (2SF)	4	MKN	140

Reaction No. 58742



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:1.61 (3SF)	3	MCN	140
2	23			lgK:2.73 (3SF)	3	MCN	140

Reaction No. 60010

Leu-1 + Ti+3 = Ti+3_Leu-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:8.55(3SF)	0	MGL	2392
2	25	0.1	Unknown	lgK:8.50(3SF)	3	MGL	11516
Note (2): Farooq O et al., J. Electroanal. Chem., 1970, 24, 464							

Reaction No. 60174							
EDTA-4 + Ti+4 = Ti+4_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	0.5	H2SO4	lgK:17.7(3SF)	0	MPL	903
2	25	0	Inf. Dilution	lgK:26.63(4SF)	1	EOR	
3	25	0.2	KNO3	lgK:22.61(4SF)	3	MVM	11516
Note (3): Duanci Z et al., Anal. Chem. (China), 1986, 650							

Reaction No. 60175							
Ti+3_EDTA-4 + H+1 = Ti+3_H+1_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15			lgK:1.7(2SF)	0	MGL	140
2	20	0.1	Unknown	lgK:2.30(3SF)	4	MGL	140
3	25	0.1	NaClO4	lgK:2.02(0.01SD)	4	MGL	903

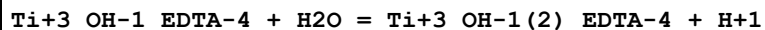
Reaction No. 60176							
Ti+4_EDTA-4 + H2O = TiO+2 + 2<H+1> + EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	0.5	H2SO4	lgK:-22.52(4SF)	0	MPL	903
Note (1): Sign changed							
2	25	0	Inf. Dilution	lgK:-24.4(3SF)	1	ELC	

Reaction No. 60205							
O2(g) + Ti(s) - 2<e-1> = Ti+4_O2-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:46.9(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-467.2(4SF)kJ	0	EOD	19339

Reaction No. 60235							
Ti+3_EDTA-4 + H2O = Ti+3_OH-1_EDTA-4 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

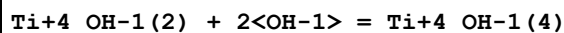
1	25	0.1	NaClO4	lgK:8.64 (3SF)	3	MGL	903
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Reaction No. 60236



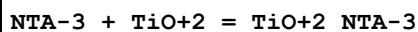
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:-11.61 (4SF)	3	MGL	903

Reaction No. 60501



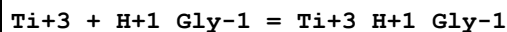
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.4 (3SF)	1	EOR	
2	25	0.1	Unknown	lgK:22.6 (3SF)	0	CRV	2556
3	25	3	Unknown	lgK:29. (2SF)	1	EOR	

Reaction No. 61486



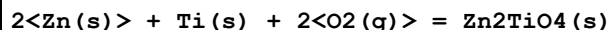
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KClO4	lgK:12.3 (0.1SD)	4	MDS	1118
2	25	0	Inf. Dilution	lgK:12.5 (3SF)	1	ESC	

Reaction No. 61595



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:1.76 (3SF)	5	CRV	552

Reaction No. 66238



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:269.5 (4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:269.5 (4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:267.7 (4SF)	4	ENS	6422
4	27	0	Inf. Dilution	lgK:267.7 (4SF)	4	ENS	6494
5	127	0	Inf. Dilution	lgK:195.8 (4SF)	4	ENS	6422
6	127	0	Inf. Dilution	lgK:195.8 (4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:152.7 (4SF)	4	ENS	6422
8	227	0	Inf. Dilution	lgK:152.7 (4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:124.1 (4SF)	4	ENS	6422

10	327	0	Inf. Dilution	lgK:124.0 (4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-367.7 (4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-367.7 (4SF)	4	EFE	6494
13	25	0	Inf. Dilution	dH:-394.9 (4SF)	4	MTD	6422
14	25	0	Inf. Dilution	dH:-394.9 (4SF)	4	MTD	6494
15	127	0	Inf. Dilution	dH:-394.6 (4SF)	4	MTD	6422
16	127	0	Inf. Dilution	dH:-394.7 (4SF)	4	MTD	6494
17	227	0	Inf. Dilution	dH:-394.1 (4SF)	4	MTD	6422
18	227	0	Inf. Dilution	dH:-394.3 (4SF)	4	MTD	6494
19	327	0	Inf. Dilution	dH:-393.6 (4SF)	4	MTD	6422
20	327	0	Inf. Dilution	dH:-393.9 (4SF)	4	MTD	6494

Reaction No. 66349							
3.5<O2(g)> + 4<Ti(s)> = Ti4O7(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:562.9 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:559.2 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:411.1 (4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:322.2 (4SF)	3	ENS	6422
5	327	0	Inf. Dilution	lgK:263.1 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-767.9 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-3213.3 (5SF) kJ	3	ENS	19339
8	25	0	Inf. Dilution	dG:-767.15 (5SF)	3	CRV	31291
9	25	0	Inf. Dilution	dH:-813.7 (4SF)	3	MTD	6422
10	127	0	Inf. Dilution	dH:-813.3 (4SF)	3	MTD	6422
11	227	0	Inf. Dilution	dH:-812.4 (4SF)	3	MTD	6422
12	327	0	Inf. Dilution	dH:-811.4 (4SF)	0	MTD	6422

Reaction No. 66350							
0.5<O2(g)> + Ti(s) = TiO(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:89.92 (4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:89.34 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:65.72 (4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:51.56 (4SF)	3	ENS	6422
5	327	0	Inf. Dilution	lgK:42.12 (4SF)	0	ENS	6422

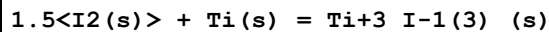
6	25	0	Inf. Dilution	dG:-122.7 (4SF)	3	EFE	6422
7	25	0	Inf. Dilution	dG:-495.0 (4SF) kJ	0	EFE	7381
8	25	0	Inf. Dilution	dG:-118.3 (4SF)	0	CRV	7382
Note (8): Low wgt for internal consistency							
9	25	0	Inf. Dilution	dG:-495.0 (4SF) kJ	0	ENS	19339
10	25	0	Inf. Dilution	dG:-489.2 (4SF) kJ	0	ENS	19339
11	25	0	Inf. Dilution	dH:-129.7 (4SF)	3	MTD	6422
12	25	0	Inf. Dilution	dH:-519.7 (4SF) kJ	0	ENS	7381
13	127	0	Inf. Dilution	dH:-129.7 (4SF)	3	MTD	6422
14	227	0	Inf. Dilution	dH:-129.5 (4SF)	3	MTD	6422
15	327	0	Inf. Dilution	dH:-129.4 (4SF)	0	MTD	6422

Reaction No. 66351							
Ti(s) + 0.5<N2(g)> = TiN(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:54.16 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:53.80 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:39.09 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:30.27 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:24.40 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-73.9 (3SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-80.8 (3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-80.7 (3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-80.7 (3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-80.5 (3SF)	4	MTD	6422

Reaction No. 66352							
2<I2(s)> + Ti(s) = Ti+4_I-1(4)_ (s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:64.94 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:64.53 (4SF)	0	ENS	6422
3	106	0	Inf. Dilution	lgK:50.88 (4SF)	2	ENS	6422
4	25	0	Inf. Dilution	dG:-88.59 (4SF)	4	EFE	6422
5	25	0	Inf. Dilution	dG:-88.8 (3SF)	4	CRV	7382
6	25	0	Inf. Dilution	dG:-88.606 (5SF)	3	CRV	31291
7	25	0	Inf. Dilution	dH:-89.8 (3SF)	3	MTD	6422

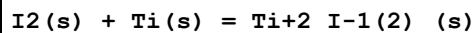
8	25	0	Inf. Dilution	dH:-89.8(3SF)	3	CRV	7382
9	25	0	Inf. Dilution	dH:-89.800(5SF)	3	CRV	31291
10	106	0	Inf. Dilution	dH:-90.0(3SF)	1	MTD	6422

Reaction No. 66353



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:55.79(4SF)	2	ENS	6422
2	27	0	Inf. Dilution	lgK:55.44(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:41.32(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:31.67(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:24.58(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-76.11(4SF)	2	EFE	6422
7	25	0	Inf. Dilution	dG:-79.849(5SF)	0	CRV	31291
8	25	0	Inf. Dilution	dH:-77.0(3SF)	2	MTD	6422
9	25	0	Inf. Dilution	dH:-80.880(5SF)	2	CRV	31291
10	127	0	Inf. Dilution	dH:-82.6(3SF)	1	MTD	6422
11	227	0	Inf. Dilution	dH:-97.7(3SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-96.9(3SF)	0	MTD	6422

Reaction No. 66354



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:45.35(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:45.06(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:33.42(4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:25.64(4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:20.02(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-61.87(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-61.877(5SF)	3	CRV	31291
8	25	0	Inf. Dilution	dH:-63.6(3SF)	4	MTD	6422
9	25	0	Inf. Dilution	dH:-63.600(5SF)	3	CRV	31291
10	127	0	Inf. Dilution	dH:-67.3(3SF)	2	MTD	6422
11	227	0	Inf. Dilution	dH:-77.4(3SF)	2	MTD	6422
12	327	0	Inf. Dilution	dH:-76.9(3SF)	0	MTD	6422

Reaction No. 66355

H ₂ (g) + Ti(s) = TiH ₂ (s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.41(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:18.25(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:11.92(4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:8.080(4SF)	3	ENS	6422
5	327	0	Inf. Dilution	lgK:5.501(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-25.11(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-19.2(3SF)	0	CRV	7382
Note (7): Low wgt for internal consistency							
8	25	0	Inf. Dilution	dG:-80.3(3SF) kJ	0	ENS	19339
Note (8): Low wgt for internal consistency							
9	25	0	Inf. Dilution	dG:-4.92(3SF) kJ	0	ENS	19339
Note (9): Low wgt for internal consistency							
10	25	0	Inf. Dilution	dH:-34.5(3SF)	4	MTD	6422
11	127	0	Inf. Dilution	dH:-35.0(3SF)	3	MTD	6422
12	227	0	Inf. Dilution	dH:-35.3(3SF)	3	MTD	6422
13	327	0	Inf. Dilution	dH:-35.5(3SF)	0	MTD	6422

Reaction No. 66356							
2<F2(g)> + Ti(s) = Ti+4_F-1(4)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:273.2(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:271.4(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:199.6(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:156.7(4SF)	4	ENS	6422
5	285.3	0	Inf. Dilution	lgK:138.8(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-372.7(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-1448.9(5SF) kJ	0	ENS	19339
Note (7): Low wgt for internal consistency							
8	25	0	Inf. Dilution	dH:-394.2(4SF)	4	MTD	6422
9	127	0	Inf. Dilution	dH:-393.4(4SF)	4	MTD	6422
10	227	0	Inf. Dilution	dH:-392.6(4SF)	4	MTD	6422
11	285.3	0	Inf. Dilution	dH:-392.0(4SF)	4	MTD	6422

Reaction No. 66357							
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1.5<F2(g)> + Ti(s) = Ti+3_F-1(3)_ (s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:238.6(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:237.0(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:174.6(4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:137.2(4SF)	3	ENS	6422
5	327	0	Inf. Dilution	lgK:112.3(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-325.5(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-1217.1(5SF) kJ	0	ENS	19339
Note (7): Low wgt for internal consistency							
8	25	0	Inf. Dilution	dH:-343.1(4SF)	4	MTD	6422
9	127	0	Inf. Dilution	dH:-342.6(4SF)	3	MTD	6422
10	227	0	Inf. Dilution	dH:-342.2(4SF)	3	MTD	6422
11	327	0	Inf. Dilution	dH:-341.8(4SF)	0	MTD	6422

Reaction No. 66358							
Ti(s) + 2<Cl2(g)> = Ti+4_Cl-1(4)_ (s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:129.2(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:128.3(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:93.39(4SF)	4	ENS	6422
4	135	0	Inf. Dilution	lgK:91.34(4SF)	4	ENS	6422
5	25	0	Inf. Dilution	dG:-176.2(4SF)	4	EFE	6422
6	25	0	Inf. Dilution	dH:-192.2(4SF)	4	MTD	6422
7	127	0	Inf. Dilution	dH:-191.0(4SF)	4	MTD	6422
8	135	0	Inf. Dilution	dH:-190.9(4SF)	4	MTD	6422

Reaction No. 66359							
1.5<Cl2(g)> + Ti(s) = Ti+3_Cl-1(3)_ (s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:114.7(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:113.9(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:82.50(4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:63.73(4SF)	3	ENS	6422
5	327	0	Inf. Dilution	lgK:51.25(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-653.5(4SF) kJ	3	CRV	6330

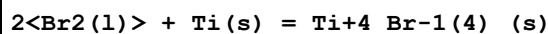
7	25	0	Inf. Dilution	dG:-156.4 (4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dG:-156.2 (4SF)	5	CRV	7382
9	25	0	Inf. Dilution	dH:-720.9 (4SF) kJ	3	CRV	6330
10	25	0	Inf. Dilution	dH:-172.5 (4SF)	4	MTD	6422
11	127	0	Inf. Dilution	dH:-172.0 (4SF)	3	MTD	6422
12	227	0	Inf. Dilution	dH:-171.5 (4SF)	3	MTD	6422
13	327	0	Inf. Dilution	dH:-171.1 (4SF)	0	MTD	6422

Reaction No. 66360							
Cl ₂ (g) + Ti(s) = Ti+2_Cl-1(2)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:81.61 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:81.05 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:58.64 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:45.22 (4SF)	3	ENS	6422
5	327	0	Inf. Dilution	lgK:36.30 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-464.4 (4SF) kJ	3	CRV	6330
7	25	0	Inf. Dilution	dG:-111.3 (4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dG:-111.0 (4SF)	5	CRV	7382
9	25	0	Inf. Dilution	dH:-513.8 (4SF) kJ	3	CRV	6330
10	25	0	Inf. Dilution	dH:-123.2 (4SF)	4	MTD	6422
11	127	0	Inf. Dilution	dH:-122.9 (4SF)	4	MTD	6422
12	227	0	Inf. Dilution	dH:-122.6 (4SF)	3	MTD	6422
13	327	0	Inf. Dilution	dH:-122.3 (4SF)	0	MTD	6422

Reaction No. 66361							
Ti(s) + C(s) = TiC(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:31.68 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:31.48 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:23.45 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:18.64 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:15.44 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-43.22 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-44.1 (3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-44.1 (3SF)	4	MTD	6422

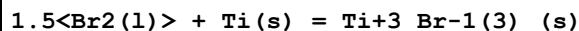
9	227	0	Inf. Dilution	dH:-44.0 (3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-43.9 (3SF)	4	MTD	6422

Reaction No. 66362



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:103.5 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:102.8 (4SF)	0	ENS	6422
3	38.4	0	Inf. Dilution	lgK:98.87 (4SF)	3	ENS	6422
4	25	0	Inf. Dilution	dG:-589.5 (4SF) kJ	3	CRV	6330
5	25	0	Inf. Dilution	dG:-141.2 (4SF)	4	EFE	6422
6	25	0	Inf. Dilution	dG:-140.9 (4SF)	4	CRV	7382
7	25	0	Inf. Dilution	dH:-616.7 (4SF) kJ	3	CRV	6330
8	25	0	Inf. Dilution	dH:-147.7 (4SF)	4	MTD	6422
9	25	0	Inf. Dilution	dH:-147.4 (4SF)	4	CRV	7382
10	38.4	0	Inf. Dilution	dH:-147.8 (4SF)	3	MTD	6422

Reaction No. 66363



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:92.08 (4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:91.49 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:66.39 (4SF)	2	ENS	6422

Note (3): 12

4	227	0	Inf. Dilution	lgK:50.85 (4SF)	3	ENS	6422
5	327	0	Inf. Dilution	lgK:40.58 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-523.8 (4SF) kJ	3	CRV	6330
7	25	0	Inf. Dilution	dG:-125.6 (4SF)	3	EFE	6422
8	25	0	Inf. Dilution	dG:-125.2 (4SF)	3	CRV	7382
9	25	0	Inf. Dilution	dH:-548.5 (4SF) kJ	3	CRV	6330
10	25	0	Inf. Dilution	dH:-131.5 (4SF)	3	MTD	6422
11	25	0	Inf. Dilution	dH:-131.1 (4SF)	3	CRV	7382
12	127	0	Inf. Dilution	dH:-142.0 (4SF)	2	MTD	6422
13	227	0	Inf. Dilution	dH:-141.4 (4SF)	2	MTD	6422
14	327	0	Inf. Dilution	dH:-140.5 (4SF)	0	MTD	6422

Reaction No. 66364

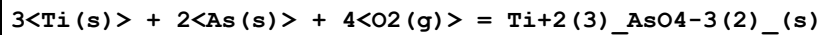
Br ₂ (l) + Ti(s) = Ti+2_Br-1(2)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:67.13(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:66.70(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:48.25(4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:36.92(4SF)	3	ENS	6422
5	327	0	Inf. Dilution	lgK:29.40(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-91.58(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-402.0(4SF)kJ	3	CRV	6330
8	25	0	Inf. Dilution	dH:-96.9(3SF)	4	MTD	6422
9	127	0	Inf. Dilution	dH:-103.9(4SF)	3	MTD	6422
10	227	0	Inf. Dilution	dH:-103.5(4SF)	3	MTD	6422
11	327	0	Inf. Dilution	dH:-103.1(4SF)	0	MTD	6422

Reaction No. 66365							
Ti(s) + 2<B(s)> = TiB ₂ (s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:56.00(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:55.65(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:41.55(4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:33.08(4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:27.43(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-76.40(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-77.4(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-77.5(3SF)	3	MTD	6422
9	227	0	Inf. Dilution	dH:-77.5(3SF)	2	MTD	6422
10	327	0	Inf. Dilution	dH:-77.6(3SF)	0	MTD	6422

Reaction No. 66366							
Ti(s) + B(s) = TiB(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.98(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:27.80(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:20.82(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:16.62(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:13.82(4SF)	4	ENS	6422

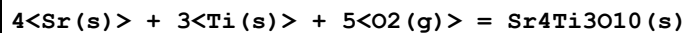
6	25	0	Inf. Dilution	dG:-38.17 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-38.3 (3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-38.4 (3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-38.4 (3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-38.4 (3SF)	4	MTD	6422

Reaction No. 66367



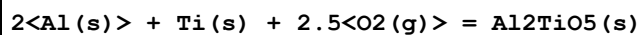
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:446.3 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:443.4 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:329.7 (4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:261.7 (4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:216.6 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-608.8 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-625.5 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-623.3 (4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-620.9 (4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-618.2 (4SF)	0	MTD	6422

Reaction No. 66392



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:939.3 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:933.2 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:687.3 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:540.0 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:441.8 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-1281. (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-1350. (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-1349. (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-1348. (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-1347. (4SF)	4	MTD	6422

Reaction No. 66481



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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1	25	0	Inf. Dilution	lgK:434.9 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:432.1 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:317.7 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:249.0 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:203.3 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-593.3 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-628.3 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-628.3 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-628.0 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-627.6 (4SF)	4	MTD	6422

Reaction No. 66516



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:373.7 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:371.3 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:273.7 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:215.1 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:176.1 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-509.8 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-536.1 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-535.6 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-535.4 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-535.5 (4SF)	4	MTD	6422

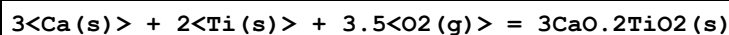
Reaction No. 66584



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:283.0 (4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:276.0 (4SF)	0	ENS	6422
3	25	0	Inf. Dilution	lgK:275.9 (4SF)	0	ENS	6494
4	27	0	Inf. Dilution	lgK:274.2 (4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:274.1 (4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:201.9 (4SF)	0	ENS	6422
7	127	0	Inf. Dilution	lgK:201.8 (4SF)	0	ENS	6494
8	227	0	Inf. Dilution	lgK:158.6 (4SF)	0	ENS	6422

9	227	0	Inf. Dilution	lgK:158.5 (4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:129.7 (4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:129.7 (4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-376.5 (4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-376.4 (4SF)	0	EFE	6494
14	25	0	Inf. Dilution	dH:-396.9 (4SF)	2	MTD	6422
15	25	0	Inf. Dilution	dH:-396.9 (4SF)	0	MTD	6494
16	127	0	Inf. Dilution	dH:-396.7 (4SF)	0	MTD	6422
17	127	0	Inf. Dilution	dH:-396.6 (4SF)	0	MTD	6494
18	227	0	Inf. Dilution	dH:-396.3 (4SF)	0	MTD	6422
19	227	0	Inf. Dilution	dH:-396.2 (4SF)	0	MTD	6494
20	327	0	Inf. Dilution	dH:-395.9 (4SF)	0	MTD	6422
21	327	0	Inf. Dilution	dH:-395.8 (4SF)	0	MTD	6494

Reaction No. 66585



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:671.2 (4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:657.2 (4SF)	0	ENS	6422
3	27	0	Inf. Dilution	lgK:652.9 (4SF)	0	ENS	6422
4	127	0	Inf. Dilution	lgK:481.0 (4SF)	0	ENS	6422
5	227	0	Inf. Dilution	lgK:377.9 (4SF)	0	ENS	6422
6	327	0	Inf. Dilution	lgK:309.3 (4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-896.5 (4SF)	0	EFE	6422
8	25	0	Inf. Dilution	dH:-944.2 (4SF)	2	MTD	6422
9	127	0	Inf. Dilution	dH:-943.6 (4SF)	0	MTD	6422
10	227	0	Inf. Dilution	dH:-942.8 (4SF)	0	MTD	6422
11	327	0	Inf. Dilution	dH:-941.9 (4SF)	0	MTD	6422

Reaction No. 66586



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:943.8 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:937.6 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:690.8 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:542.8 (4SF)	4	ENS	6422

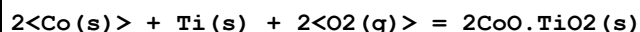
5	327	0	Inf. Dilution	lgK:444.3 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-1287. (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-1356. (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-1355. (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-1353.5 (5SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-1352. (4SF)	4	MTD	6422

Reaction No. 66587							
Ca(s) + Si(s) + Ti(s) + 2.5<O2(g)> = CaO.SiO2.TiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:431.3 (4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:430.0 (4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:428.5 (4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:427.2 (4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:315.2 (4SF)	2	ENS	6422
6	127	0	Inf. Dilution	lgK:314.2 (4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:247.2 (4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:246.4 (4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:201.9 (4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:201.2 (4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-588.4 (4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-586.79 (5SF)	5	EFE	6482
13	25	0	Inf. Dilution	dG:-586.7 (4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dH:-622.2 (4SF)	4	MTD	6422
15	25	0	Inf. Dilution	dH:-620.61 (5SF)	5	EFE	6482
16	25	0	Inf. Dilution	dH:-620.6 (4SF)	4	MTD	6494
17	127	0	Inf. Dilution	dH:-622.1 (4SF)	1	MTD	6422
18	127	0	Inf. Dilution	dH:-620.5 (4SF)	1	MTD	6494
19	227	0	Inf. Dilution	dH:-621.7 (4SF)	0	MTD	6422
20	227	0	Inf. Dilution	dH:-620.2 (4SF)	0	MTD	6494
21	327	0	Inf. Dilution	dH:-621.3 (4SF)	0	MTD	6422
22	327	0	Inf. Dilution	dH:-619.8 (4SF)	0	MTD	6494

Reaction No. 66612							
Cd(s) + Ti(s) + 1.5<O2(g)> = CdTiO3(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

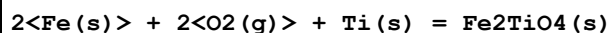
1	25	0	Inf. Dilution	lgK:200.7 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:199.4 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:145.8 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:113.7 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:92.31 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-273.8 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-294.1 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-293.9 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-293.7 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-294.9 (4SF)	4	MTD	6422

Reaction No. 66642



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:235.8 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:234.2 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:171.3 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:133.6 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:108.5 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-321.7 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-345.6 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-345.2 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-344.7 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-344.3 (4SF)	4	MTD	6422

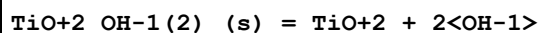
Reaction No. 66730



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:248.4 (4SF)	0	ENS	6422
2	25	0	Inf. Dilution	lgK:245.3 (4SF)	0	ENS	6494
3	27	0	Inf. Dilution	lgK:246.8 (4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:243.6 (4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:180.8 (4SF)	0	ENS	6422
6	127	0	Inf. Dilution	lgK:178.7 (4SF)	0	ENS	6494
7	227	0	Inf. Dilution	lgK:141.3 (4SF)	0	ENS	6422
8	227	0	Inf. Dilution	lgK:139.7 (4SF)	0	ENS	6494

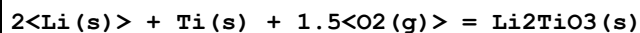
9	327	0	Inf. Dilution	lgK:115.0 (4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:113.8 (4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-338.9 (4SF)	0	EFE	6422
12	25	0	Inf. Dilution	dG:-334.6 (4SF)	2	EFE	6494
13	25	0	Inf. Dilution	dG:-1399.9 (5SF) kJ	3	EOD	12843
14	25	0	Inf. Dilution	dH:-362.2 (4SF)	0	MTD	6422
15	25	0	Inf. Dilution	dH:-357.0 (4SF)	2	MTD	6494
16	25	0	Inf. Dilution	dH:-1493.8 (5SF) kJ	3	EOD	12843
17	27	0	Inf. Dilution	dH:-362.1 (4SF)	0	MTD	6422
18	127	0	Inf. Dilution	dH:-361.9 (4SF)	0	MTD	6422
19	127	0	Inf. Dilution	dH:-356.6 (4SF)	0	MTD	6494
20	227	0	Inf. Dilution	dH:-361.5 (4SF)	0	MTD	6422
21	227	0	Inf. Dilution	dH:-356.2 (4SF)	0	MTD	6494
22	327	0	Inf. Dilution	dH:-361.1 (4SF)	0	MTD	6422
23	327	0	Inf. Dilution	dH:-355.8 (4SF)	0	MTD	6494

Reaction No. 66850



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	3.5	Unknown	lgK:-30.42 (4SF)	3	EOD	20328
2	25	0	Inf. Dilution	lgK:-32.8 (3SF)	4	ENS	773
3	25	1	Unknown	lgK:-31.9 (3SF)	4	ENS	773

Reaction No. 66958



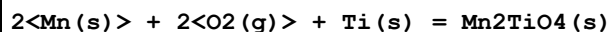
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:276.8 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:275.0 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:202.2 (4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:158.5 (4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:129.3 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-377.6 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-399.3 (4SF)	3	MTD	6422
8	127	0	Inf. Dilution	dH:-399.4 (4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-400.8 (4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-400.7 (4SF)	0	MTD	6422

Reaction No. 66975							
$\text{Mg(s)} + 2.5\text{O}_2\text{(g)} + 2\text{Ti(s)} = \text{MgO} \cdot 2\text{TiO}_2\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:415.0 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:412.3 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:303.1 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:237.6 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:193.9 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-566.1 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-599.8 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-599.6 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-599.3 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-598.8 (4SF)	4	MTD	6422

Reaction No. 66997							
$\text{Mn(s)} + 1.5\text{O}_2\text{(g)} + \text{Ti(s)} = \text{MnTiO}_3\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:224.1 (4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:244.4 (4SF)	0	ENS	6494
3	27	0	Inf. Dilution	lgK:222.7 (4SF)	4	ENS	6422
4	27	0	Inf. Dilution	lgK:222.9 (4SF)	4	ENS	6494
5	127	0	Inf. Dilution	lgK:163.6 (4SF)	4	ENS	6422
6	127	0	Inf. Dilution	lgK:163.7 (4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:128.1 (4SF)	4	ENS	6422
8	227	0	Inf. Dilution	lgK:128.3 (4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:104.5 (4SF)	4	ENS	6422
10	327	0	Inf. Dilution	lgK:104.6 (4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-305.8 (4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-306.1 (4SF)	4	EFE	6494
13	25	0	Inf. Dilution	dH:-324.7 (4SF)	4	MTD	6422
14	25	0	Inf. Dilution	dH:-322.7 (4SF)	4	MTD	6494
15	127	0	Inf. Dilution	dH:-324.5 (4SF)	4	MTD	6422
16	127	0	Inf. Dilution	dH:-324.9 (4SF)	4	MTD	6494
17	227	0	Inf. Dilution	dH:-324.2 (4SF)	4	MTD	6422
18	227	0	Inf. Dilution	dH:-324.6 (4SF)	4	MTD	6494
19	327	0	Inf. Dilution	dH:-323.8 (4SF)	4	MTD	6422

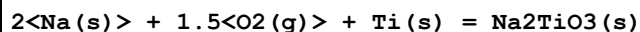
20	327	0	Inf. Dilution	dH:-324.3(4SF)	4	MTD	6494
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Reaction No. 66998



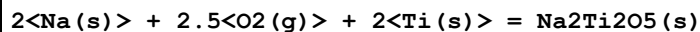
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:289.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:287.2(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:211.0(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:165.4(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:135.0(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-394.4(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-418.2(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-417.9(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-417.5(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-417.1(4SF)	4	MTD	6422

Reaction No. 67051



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:255.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:253.5(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:185.9(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:145.2(4SF)	4	ENS	6422
5	287	0	Inf. Dilution	lgK:127.8(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-348.1(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-370.8(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-372.1(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-372.0(4SF)	4	MTD	6422
10	287	0	Inf. Dilution	dH:-371.9(4SF)	4	MTD	6422

Reaction No. 67052



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:418.6(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:415.9(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:305.3(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:238.8(4SF)	4	ENS	6422

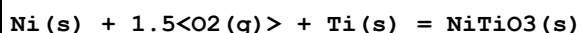
5	327	0	Inf. Dilution	lgK:194.5 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-571.1 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-607.0 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-608.4 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-608.4 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-608.4 (4SF)	4	MTD	6422

Reaction No. 67053



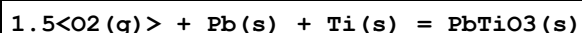
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:574.2 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:570.4 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:418.9 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:327.9 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:267.2 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-783.3 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-831.7 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-833.0 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-833.0 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-832.9 (4SF)	4	MTD	6422

Reaction No. 67107



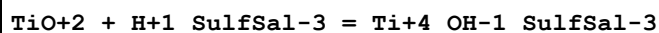
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:195.9 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:194.6 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:142.3 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:110.9 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:90.01 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-267.2 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-287.4 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-287.2 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-286.9 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-286.7 (4SF)	4	MTD	6422

Reaction No. 67149



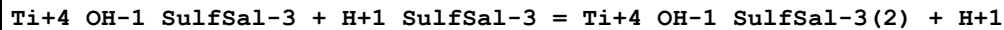
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:194.8 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:193.5 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:141.3 (4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:110.1 (4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:89.29 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-265.7 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-286.5 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-286.2 (4SF)	3	MTD	6422
9	227	0	Inf. Dilution	dH:-285.8 (4SF)	2	MTD	6422
10	327	0	Inf. Dilution	dH:-285.3 (4SF)	0	MTD	6422

Reaction No. 69115



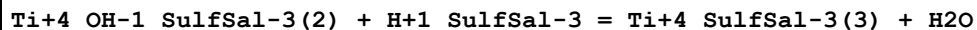
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	21	0.04	NaClO4	lgK:4.1 (2SF)	6	MPL	5987
2	25	0	Inf. Dilution	lgK:4.2 (2SF)	1	ESC	

Reaction No. 69116



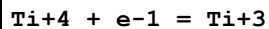
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	21	0.6	NaClO4	lgK:-0.2 (1SF)	6	MPL	5987
2	25	0	Inf. Dilution	lgK:-0.2 (1SF)	1	ESC	

Reaction No. 69117



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	21	0.6	NaClO4	lgK:1.7 (2SF)	6	MPL	5987
2	25	0	Inf. Dilution	lgK:1.7 (2SF)	1	ESC	

Reaction No. 70247



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.9 (2SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.2 (1SF)	3	ENS	7276
3	25	0	Inf. Dilution	E:0.202 (0.002SD)	5	CRV	23624
4	25	0.1:3	HCl	E:0.11 (2SF)	0	MPT	20357

Reaction No. 70521							
Ti(s) = Ti(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:425.1(4SF) kJ	2	EFE	7381
2	25	0	Inf. Dilution	dG:428.4(4SF) kJ	6	CRV	8746
3	25	0	Inf. Dilution	dG:428.40(3.000SD) kJ	6	CRV	14923
4	25	0	Inf. Dilution	dH:469.9(4SF) kJ	3	ENS	7381
5	25	0	Inf. Dilution	dH:473.(3.MD) kJ	6	CRV	8746
6	25	0	Inf. Dilution	dH:473.00(3.000SD) kJ	6	CRV	14923
7	25	0	Inf. Dilution	Cp:24.430(0.015SD) J/K	5	CRV	27292

Reaction No. 70942							
TiO2(Anatase,s) + 4<H+1> + 2<e-1> = Ti+2 + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-7.4(2SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.502(3SF)	0	CRV	6330

Reaction No. 73594							
3<S(s)> + 2<Ti(s)> = Ti+3(2)_S-2(3)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-784.6(4SF) kJ	3	EFE	19339

Reaction No. 73596							
0.5<H2(g)> + 1.5<O2(g)> + Ti(s) + e-1 = HTiO3-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-955.9(4SF) kJ	3	EOD	19339

Reaction No. 73597							
1.5<O2(g)> + 2<Ti(s)> = Ti2O3(aq.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1388.(4SF) kJ	3	ENS	19339

Reaction No. 73598							
H2(g) + 1.5<O2(g)> + Ti(s) = TiO+2_OH-1(2)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:189.3(4SF)	1	ELC	

2	25	0	Inf. Dilution	dG:-1058.5 (5SF) kJ	0	ENS	19339
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Reaction No. 73599							
$O_2(g) + Ti(s) = TiO_2(s)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-888.4 (4SF) kJ	0	ENS	19339
Note (1): No phase. Unreliable dG seemingly between Anatase & Rutile							
2	25	0	Inf. Dilution	dG:-212.47 (5SF)	2	CRV	31291
Note (2): this refers to rutile							

Reaction No. 73600							
$0.5H_2(g) + Ti(s) = TiH(s)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-3.59 (3SF) kJ	3	ENS	19339

Reaction No. 73601							
$2H_2(g) + 2.5O_2(g) + Ti(s) = TiO_3 \cdot 2H_2O(s)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1173. (4SF) kJ	3	ENS	19339

Reaction No. 73602							
$3F_2(g) + Ti(s) + 2e^- = TiF_3(l)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:375.4 (4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-2118.4 (5SF) kJ	0	ENS	19339

Reaction No. 73603							
$2Cl_2(g) + Ti(s) = TiCl_4(l)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-737.2 (4SF) kJ	3	CRV	6330
2	25	0	Inf. Dilution	dG:-176.2 (4SF)	4	CRV	7382
3	25	0	Inf. Dilution	dH:-804.2 (4SF) kJ	3	CRV	6330
4	25	0	Inf. Dilution	dH:-192.2 (4SF)	4	CRV	7382

Reaction No. 73604							
$TiO_2(Rutile, s) + 2H_2O = Ti(OH)_4(l)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:-14.5 (3SF)	1	ELC	
2	200	0	Inf. Dilution	lgK:-7.56 (3SF)	3	CRV	20328
3	300	0	Inf. Dilution	lgK:-7.48 (3SF)	0	CRV	20328
4	25	0	Inf. Dilution	dG:51.4 (0.23SD) kJ	0	MSL	16743
Note (4): Incorrect conditions?							

Reaction No. 73605							
Ti+4_OH-1(4) + H2O = Ti+4_OH-1(5) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:70.46 (1.18SD) kJ	6	MSL	16743

Reaction No. 73606							
TiO2(Rutile,s) + H2O + H+1_PO4-3 = Ti+4_OH-1(3)_PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:57.74 (4SF) kJ	0	MSL	16743
Note (1): Source reports as a diff. reaction but species differ only in H2O							
2	25	0	Inf. Dilution	dG:34.43 (0.34SD) kJ	2	MSL	16743

Reaction No. 73607							
TiO2(Rutile,s) + 2<H2O> + PO4-3 = Ti+4_OH-1(4)_PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:51.01 (4SF) kJ	6	MSL	16743

Reaction No. 73608							
Ti(s) - 4<e-1> = Ti+4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:57.4 (3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-354.18 (3.85SD) kJ	0	CRV	16743
3	25	0	Inf. Dilution	dG:-84.65 (0.92SD)	0	EFE	20328
4	25	0	Inf. Dilution	dH:-457.00 (4.8SD) kJ	0	CRV	16743

Reaction No. 73609							
0.5<H2(g)> + 0.5<O2(g)> + Ti(s) - 3<e-1> = Ti+4_OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:100.0 (4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-614.00 (3.85SD) kJ	0	CRV	16743

Reaction No. 73610							
$\text{H}_2(\text{g}) + \text{O}_2(\text{g}) + \text{Ti}(\text{s}) - 2\text{e}^- = \text{Ti}^{+4}\text{OH}^-1(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 142.0(4\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\text{dG}: -862.16(5\text{SF}) \text{ kJ}$	0	CRV	16743
3	25	0	Inf. Dilution	$\text{dH}: -976.02(3.3\text{SD}) \text{ kJ}$	0	CRV	16743

Reaction No. 73611							
$1.5\text{H}_2(\text{g}) + 1.5\text{O}_2(\text{g}) + \text{Ti}(\text{s}) - \text{e}^- = \text{Ti}^{+4}\text{OH}^-1(3)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 183.6(4\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\text{dG}: -1086.20(6\text{SF}) \text{ kJ}$	0	CRV	16743

Reaction No. 73612							
$2.5\text{H}_2(\text{g}) + 2.5\text{O}_2(\text{g}) + \text{Ti}(\text{s}) + \text{e}^- = \text{Ti}^{+4}\text{OH}^-1(5)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 253.6(4\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\text{dG}: -1479.20(6\text{SF}) \text{ kJ}$	0	MSL	16743
3	25	0	Inf. Dilution	$\text{dH}: -1733.90(6\text{SF}) \text{ kJ}$	2	MSL	16743

Reaction No. 73617							
$\text{Ti}^{+4} + \text{H}_2\text{O} = \text{TiO}_2 + 2\text{H}^+$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 1.7(2\text{SF})$	1	ELC	
2	25	0.1:3	HCl	$\lg K: 1.4(2\text{SF})$	0	MPT	20357

Reaction No. 73618							
$\text{Ti}^{+3} + 2\text{OH}^- = \text{Ti}^{+3}\text{OH}^-1(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18	1	CsCl	$\lg K: 22.57(4\text{SF})$	4	MSL	20361
2	25	0	Inf. Dilution	$\lg K: 24.0(3\text{SF})$	2	EAE	

Reaction No. 73619							
$\text{Ti}^{+3} + 3\text{OH}^- = \text{Ti}^{+3}\text{OH}^-1(3)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18	1	CsCl	$\lg K: 32.32(4\text{SF})$	4	MSL	20361
2	25	0	Inf. Dilution	$\lg K: 34.0(3\text{SF})$	2	EAE	

Reaction No. 73620							
$\text{Ti}+3 + 3\langle\text{OH}-1\rangle = \text{Ti}+3_{\text{OH}-1(3)}_{\text{(s)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18	1	CsCl	lgK:37.29(4SF)	0	MSL	20361
2	25	0	Inf. Dilution	lgK:40.0(3SF)	1	ELC	

Reaction No. 73621							
$3\langle\text{TiO}+2\rangle + \text{H}_2\text{O} = \text{Ti}_3\text{O}_4+4 + 2\langle\text{H}+1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	4	HClO ₄	lgR:-0.42(2SF)	6	MMR	20330

Reaction No. 73622							
$3\langle\text{TiO}+2\rangle + 2\langle\text{H}_2\text{O}\rangle = \text{Ti}_3\text{O}_5\text{H}_3+5 + \text{H}+1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	4	HClO ₄	lgR:0.215(3SF)	6	MMR	20330

Reaction No. 73623							
$4\langle\text{TiO}+2\rangle + 2\langle\text{H}_2\text{O}\rangle = \text{Ti}_4\text{O}_6\text{H}_4+8$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	4	HClO ₄	lgK:0.364(3SF)	6	MMR	20330

Reaction No. 73624							
$\text{Ti}+4 + 4\langle\text{OH}-1\rangle = \text{Ti}+4_{\text{OH}-1(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:56.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:58.68(4SF)	0	CRV	20328
3	25	1	NaClO ₄	lgK:53.3(3SF)	0	MDS	5398
4	25	1	Unknown	lgK:53.3(3SF)	0	EOD	20328
Note (4): Sig. lower value attributed to effect of an organic solvent							

Reaction No. 73625							
$\text{Ti}+4_{\text{OH}-1(2)} + 2\langle\text{OH}-1\rangle = \text{Ti}+4_{\text{OH}-1(4)}_{\text{(s)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	3.5	KCl	lgK:30.4(3SF)	0	MSL	5398
2	25	0	Inf. Dilution	lgK:23.7(3SF)	1	ELC	

Reaction No. 73626							
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Ti+4 + H2O = Ti+4_OH-1 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.0 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-0.3 (1SF)	0	CRV	23624
3	25	1	Unknown	lgK:-0.16 (2SF)	3	EOD	20328

Reaction No. 73627							
Ti+4 + 2<H2O> = Ti+4_OH-1 (2) + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.4 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:1.38 (3SF)	3	CRV	23624
3	25	1	Unknown	lgK:-0.59 (2SF)	3	EOD	20328

Reaction No. 73628							
Ti+4 + 3<H2O> = Ti+4_OH-1 (3) + 3<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.5 (2SF)	1	ELC	
2	25	1	Unknown	lgK:-0.92 (2SF)	0	EOD	20328

Reaction No. 73629							
Ti+4 + 4<H2O> = Ti+4_OH-1 (4) + 4<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.81 (2SF)	1	ELC	
2	25	1	Unknown	lgK:-1.05 (3SF)	0	EOD	20328

Reaction No. 73630							
Ti+4 + 2<OH-1> = Ti+4_OH-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:29.5 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:35.17 (4SF)	0	CRV	20328

Reaction No. 73631							
Ti+4 + 3<OH-1> = Ti+4_OH-1 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:43.5 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:46.68 (4SF)	0	CRV	20328

Reaction No. 74032							
$2\text{TiO}_2(\text{s}) + 2\text{e}^- + 2\text{H}^+ = \text{Ti}_2\text{O}_3(\text{s}) + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -18.65(4\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$E: -0.556(3\text{SF})$	0	CRV	22551

Reaction No. 74033							
$\text{Ti}(\text{s}) + 2\text{e}^- + 2\text{H}^+ = \text{TiH}_2(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 18.41(4\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$E: -0.45(2\text{SF})$	0	CRV	22551

Reaction No. 74034							
$\text{Ti}_4\text{F}_{-1}(\text{g}) + 4\text{e}^- = \text{Ti}(\text{s}) + 6\text{F}_{-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$E: -1.19(3\text{SF})$	5	CRV	22551
Note (1): One of few Ti rxns to be reversible							

Reaction No. 74035							
$\text{Ti}_4\text{F}_{-1}(\text{g}) + 4\text{e}^- = \text{Ti}(\text{s}) + 4\text{F}_{-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -76.3(3\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$E: -0.89(2\text{SF})$	0	CRV	22551

Reaction No. 74036							
$2\text{TiO}_2(\text{s}) + 2\text{e}^- + \text{H}_2\text{O} = \text{Ti}_2\text{O}_3(\text{s}) + 2\text{OH}^-$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$E: -1.38(3\text{SF})$	5	CRV	22551

Reaction No. 74155							
$\text{TiO}_2(\text{s}) + 4\text{H}^+ + 2\text{e}^- = \text{Ti}^{2+} + 2\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -17.5(3\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$E: -0.502(3\text{SF})$	0	CRV	22519

Reaction No. 75070							
$\text{Mg}_2\text{TiO}_4(\text{s}) + 8\text{H}^+ + \text{e}^- = 2\text{Mg}^{2+} + \text{Ti}^{3+} + 4\text{H}_2\text{O}$							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.2 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:21.144 (5SF)	0	CRV	32224

Reaction No. 75071							
$\text{MgO} \cdot 2\text{TiO}_2(\text{s}) + 10\text{H}^+ + 2\text{e}^- = \text{Mg}^{2+} + 2\text{Ti}^{3+} + 5\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-5.0 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-18.880 (5SF)	0	CRV	32224

Reaction No. 75073							
$\text{MgTiO}_3(\text{meta}, \text{s}) + 6\text{H}^+ + \text{e}^- = \text{Mg}^{2+} + \text{Ti}^{3+} + 3\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.567 (4SF)	4	CRV	32224

Reaction No. 75074							
$\text{MgTiO}_3(\text{geik}, \text{s}) + 6\text{H}^+ + \text{e}^- = \text{Mg}^{2+} + \text{Ti}^{3+} + 3\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.6 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-1.547 (4SF)	0	CRV	32224

Reaction No. 75596							
$2\text{Br}_2(\text{l}) + \text{Ti}(\text{s}) = \text{Ti}^{4+}\text{Br}_4(\text{g})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-568.2 (4SF) kJ	3	CRV	6330
2	25	0	Inf. Dilution	dH:-549.4 (4SF) kJ	3	CRV	6330

Reaction No. 75676							
$\text{Ti}^{3+}_{\text{bAla-1}} + \text{bAla-1} = \text{Ti}^{3+}_{\text{bAla-1}(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.0 (3SF)	1	EAE	
2	25	1	NaClO4	lgK:11.62 (4SF)	3	MEF	11516
Note (2): Yakovlev Y et al., Zhur. Neorg. Khim., 1977, 22, 87							

Reaction No. 75677							
$\text{Ti}^{3+} + 2\text{bAla-1} = \text{Ti}^{3+}_{\text{bAla-1}(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:25.8(3SF)	1	EAE	
2	25	1	NaClO4	lgK:24.90(4SF)	3	MEF	11516
Note (2): Yakovlev Y et al., Zhur. Neorg. Khim., 1977, 22, 87							

Reaction No. 75951							
Oxalic-2 + Ti+4 = Ti+4_Oxalic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.924(4SF)	1	EOR	
2	25	0.2	NaNO3	lgK:6.26(3SF)	1	MVM	11516
Note (2): Maluka L et al., Zhur. Neorg. Khim., 1980, 25, 1809							
3	25	1	NaClO4	lgK:5.18(3SF)	1	MVM	11516
Note (3): Turyan Y et al., Zhur. Obsh. Khim., 1979, 49, 246							

Reaction No. 75952							
Ti+4_Oxalic-2 + Oxalic-2 = Ti+4_Oxalic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.2	NaNO3	lgK:4.52(3SF)	5	MVM	11516
Note (1): Maluka L et al., Zhur. Neorg. Khim., 1980, 25, 1809							

Reaction No. 76126							
Ti+3_SO4-2 + SO4-2 = Ti+3_SO4-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18	3	Unknown	lgK:0.90(2SF)	5	MIX	20375
2	25	0	Inf. Dilution	lgK:1.16(3SF)	2	EAE	

Reaction No. 76151							
Ti+3 + SCN-1 = Ti+3_SCN-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:1.38(3SF)	5	MVM	11516
Note (1): Turyan Y et al., Zhur. Obshch. Khim., 1980, 50, 1917							

Reaction No. 76152							
Ti+3_SCN-1 + SCN-1 = Ti+3_SCN-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:0.40(2SF)	5	MVM	11516
Note (1): Turyan Y et al., Zhur. Obshch. Khim., 1980, 50, 1917							

Reaction No. 76153							
Ti+3 + 2<SCN-1> = Ti+3_SCN-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:1.78(3SF)	5	MVM	11516
Note (1): Turyan Y et al., Zhur. Obshch. Khim., 1980, 50, 1917							

Reaction No. 76154							
Ti+4 + SCN-1 = Ti+4_SCN-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:0.45(2SF)	5	MSP	19238

Reaction No. 76201							
Ti+4 + CDTA-4 = Ti+4_CDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	NaClO4	lgK:18.23(4SF)	5	MSP	11516
Note (1): Nenova P et al., Zhur. Neorg. Khim., 1972, 17, 1856							
2	25	0	Inf. Dilution	lgK:18.5(3SF)	1	ESC	

Reaction No. 78000							
EDTA-4 + Ti+3 + 2<H+1> = Ti+3_H+1(2)_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:26.84(4SF)	1	ELC	

Reaction No. 78068							
2<Citric-3> + Ti+4 + 2<OH-1> + H+1 = Ti+4_H+1_OH-1(2)_Citric-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:53.18(4SF)	1	ELC	

Reaction No. 78100							
Gly-1 + Ti+3 + OH-1 = Ti+3_Gly-1_OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.68(4SF)	1	ELC	

Reaction No. 78276							
EDTA-4 + TiO+2 + H+1 = TiO+2_H+1_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.94(4SF)	1	ELC	

Reaction No. 78327							
$\text{Citric-3} + \text{TiO}_2 + 2\text{H}^+ = \text{TiO}_2\text{H}_2\text{Citric-3}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.5(4SF)	1	ELC	

Reaction No. 78393							
$2\text{Citric-3} + \text{TiO}_2 + 4\text{H}^+ = \text{TiO}_2\text{H}_4\text{Citric-3(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.33(4SF)	1	ELC	

Reaction No. 78580							
$2\text{Oxalic-2} + \text{TiO}_2 + \text{Pb}^{2+} = \text{Pb}_2\text{TiO}_2\text{Oxalic-2(2)(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.51(4SF)	1	ELC	

Reaction No. 78631							
$\text{EDTA-4} + \text{Ti}^{3+} + \text{H}^+ = \text{TiHEDTA-4}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:26.08(4SF)	1	ELC	

Reaction No. 78640							
$\text{EDTA-4} + \text{Ti}^{3+} + \text{OH}^- = \text{TiOHEDTA-4}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:46.05(4SF)	1	ELC	

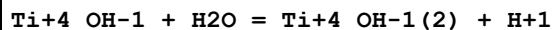
Reaction No. 78641							
$\text{EDTA-4} + \text{Ti}^{3+} + 2\text{OH}^- = \text{Ti(OH)}_2\text{EDTA-4}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:47.81(4SF)	1	ELC	

Reaction No. 79431							
$2\text{Oxalic-2} + \text{Ti}^{4+} = \text{Ti(Ox)}_2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.45(4SF)	1	ELC	

Reaction No. 80496							
$0.5\text{Ti}_2\text{O}_3(\text{s}) + 3\text{H}^+ = \text{Ti}^{3+} + 1.5\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

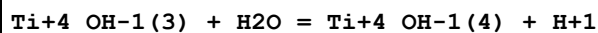
1	25	0	Inf. Dilution	lgK:-1.9 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-2.95 (3SF)	0	EDH	5493
3	25	3.5	NaClO4/m	lgK:-2.34 (3SF)	3	EOD	5493

Reaction No. 80579



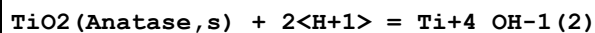
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.4 (1SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:3.26 (3SF)	0	MSL	31297
3	25	0	Inf. Dilution	dH:5.0 (2SF) kJ	2	MTD	31297

Reaction No. 80580



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.7 (1SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-3.07 (3SF)	0	MSL	31297
3	25	0	Inf. Dilution	dH:-4.0 (2SF) kJ	2	MTD	31297

Reaction No. 80581



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.2 (2SF)	2	EMN	31297
2	25	0	Inf. Dilution	lgK:-2.46 (3SF)	2	MSL	31297
3	80	0.5	HClO4	lgK:-3.60 (3SF)	2	MSL	31297

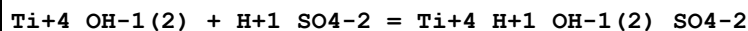
Note (3): Method 1, Table 6, Ziemniak approach

4	80	0.5	HClO4	lgK:-2.92 (3SF)	2	MSL	31297
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Note (4): Method 2, Table 6, Helgeson approach

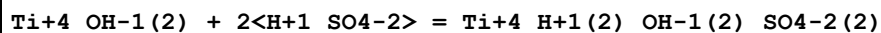
5	25	0	Inf. Dilution	dH:-13.0 (3SF) kJ	2	MTD	31297
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Reaction No. 80582



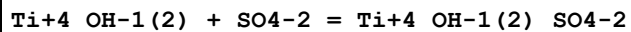
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.04 (1SF)	2	MSL	31297
2	25	0	Inf. Dilution	dH:28.1 (3SF) kJ	2	MTD	31297

Reaction No. 80583



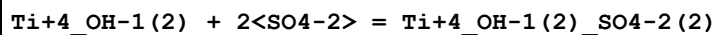
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.04 (1SF)	2	MSL	31297
2	25	0	Inf. Dilution	dH:63.0 (3SF) kJ	2	MTD	31297

Reaction No. 80584



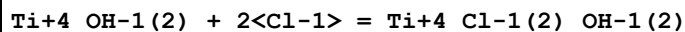
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.1 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:0.53 (2SF)	0	MSL	31297
3	25	0	Inf. Dilution	dH:33.3 (3SF) kJ	2	MTD	31297

Reaction No. 80585



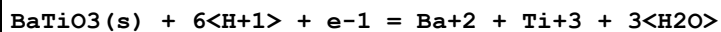
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.08 (3SF)	2	MSL	31297
2	25	0	Inf. Dilution	dH:66.5 (3SF) kJ	2	MTD	31297

Reaction No. 80586



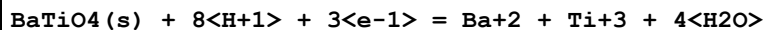
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.11 (3SF)	2	MSL	31297
2	25	0	Inf. Dilution	dH:12.0 (3SF) kJ	2	MTD	31297

Reaction No. 81331



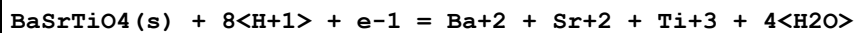
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.600 (4SF)	4	CRV	32224

Reaction No. 81332



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-55.087 (5SF)	4	CRV	32224

Reaction No. 81333



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:36.855 (5SF)	4	CRV	32224

Reaction No. 81344							
$\text{CaO.TiO}_2(\text{s}) + 6\langle\text{H}+1\rangle + \text{e}-1 = \text{Ca}+2 + \text{Ti}+3 + 3\langle\text{H}_2\text{O}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -0.171 (3\text{SF})$	4	CRV	32224

Reaction No. 81345							
$3\text{CaO}.2\text{TiO}_2(\text{s}) + 14\langle\text{H}+1\rangle + 2\langle\text{e}-1\rangle = 3\langle\text{Ca}+2\rangle + 2\langle\text{Ti}+3\rangle + 7\langle\text{H}_2\text{O}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 32.919 (5\text{SF})$	4	CRV	32224

Reaction No. 81348							
$\text{CaO.SiO}_2.\text{TiO}_2(\text{s}) + 6\langle\text{H}+1\rangle + \text{e}-1 = \text{Ca}+2 + \text{Ti}+3 + \text{H}+1(2)_{\text{SiH}_2\text{O}_4-2} + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -1.4 (2\text{SF})$	1	ELC	32224
2	25	0	Inf. Dilution	$\lg K: -9.140 (4\text{SF})$	0	CRV	32224

Reaction No. 81432							
$\text{FeTiO}_3(\text{s}) + 6\langle\text{H}+1\rangle + \text{e}-1 = \text{Fe}+2 + \text{Ti}+3 + 3\langle\text{H}_2\text{O}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -0.8 (1\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\lg K: -10.436 (5\text{SF})$	0	CRV	32224

Reaction No. 81615							
$\text{SrTiO}_3(\text{s}) + 6\langle\text{H}+1\rangle + \text{e}-1 = \text{Sr}+2 + \text{Ti}+3 + 3\langle\text{H}_2\text{O}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 5.8 (2\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\lg K: -1.44 (3\text{SF})$	0	CRV	32224

Reaction No. 81616							
$\text{SrTiO}_4(\text{s}) + 8\langle\text{H}+1\rangle + 3\langle\text{e}-1\rangle = \text{Sr}+2 + \text{Ti}+3 + 4\langle\text{H}_2\text{O}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -63.31 (4\text{SF})$	4	CRV	32224

Reaction No. 81621							
$\text{Ti}+3_{\text{Br}-1(3)}(\text{s}) = \text{Ti}+3 + 3\langle\text{Br}-1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 24.1 (3\text{SF})$	1	ELC	

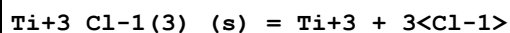
2	25	0	Inf. Dilution	lgK:8.86(3SF)	0	CRV	32224
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Reaction No. 81622



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:30.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:23.79(4SF)	0	CRV	32224

Reaction No. 81623



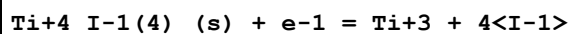
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.7(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:8.52(3SF)	0	CRV	32224

Reaction No. 81624



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.1(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:18.82(4SF)	0	CRV	32224

Reaction No. 81625



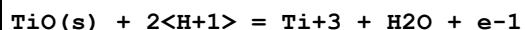
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:32.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:25.48(4SF)	0	CRV	32224

Reaction No. 81626



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:42.9(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:40.12(4SF)	0	CRV	32224

Reaction No. 81627



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.9(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:5.81(3SF)	0	CRV	32224

Reaction No. 81628							
$\text{TiO}_2(\text{Rutile},s) + 4\text{H}^+ + e^- = \text{Ti}^{3+} + 2\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-11.4 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-18.55 (4SF)	0	CRV	32224

Reaction No. 81629							
$\text{TiO}_2(\text{Anatase},s) + 4\text{H}^+ + e^- = \text{Ti}^{3+} + 2\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.2 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-17.47 (4SF)	0	CRV	32224

Reaction No. 81630							
$\text{Ti}_2\text{O}_3(s) + 6\text{H}^+ = 2\text{Ti}^{3+} + 3\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.9 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-18.20 (4SF)	0	CRV	32224

Reaction No. 81631							
$\text{Ti}_3\text{O}_5(s) + 10\text{H}^+ + e^- = 3\text{Ti}^{3+} + 5\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-14.1 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-35.70 (4SF)	0	CRV	32224

Reaction No. 81632							
$\text{Ti}_4\text{O}_7(s) + 14\text{H}^+ + 2e^- = 4\text{Ti}^{3+} + 7\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-26.7 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-55.35 (4SF)	0	CRV	32224

Reaction No. 81706							
$\text{Zn}_2\text{TiO}_4(s) + 8\text{H}^+ + e^- = 2\text{Zn}^{2+} + \text{Ti}^{3+} + 4\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.5 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:3.22 (3SF)	0	CRV	32224

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CMN	Several experimental values (crit.)
CRV	Critical review
CSM	Smith & Martell Vols 1-6 (crit.)
EAE	Automated extrapolation
EDH	Debye-Hueckel corrections
EES	Personal estimate
EFE	Free Energy relations
EIU	Critical IUPAC review (estim.)
ELC	Linear Combination of Reactions
EMN	Several experimental values (estim.)
ENS	Not specified
EOD	Calculated from data of others
EOR	Other Reaction Constants
ESC	Standard conditions anchor
MAX	Anion exchange
MCE	Cation exchange
MCL	Calorimetry

MCN	Conductivity
MDS	Distribution between two phases
MEF	Electromotoric force, not specified
MFP	Freezing point
MGL	Glass electrode
MIS	Ion selective electrode
MIX	Ion exchange
MKN	Rate of reaction
MMR	Nuclear magnetic resonance
MPH	pH method, not specified
MPL	Polarography
MPS	Potentiometry and Spectrophotometry
MPT	Potentiometry
MQH	Quinhydrone electrode
MSC	Miscellaneous
MSL	Solubility
MSP	Spectroscopy
MTD	Temperature difference
MVM	Voltametric Methods

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